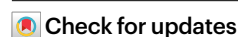


Akkermansia muciniphila protects mice against an emerging tick-borne viral pathogen

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Severe fever with thrombocytopenia syndrome (SFTS) is an emerging tick-borne disease caused by a *phlebovirus* in the Bunyaviridae family. Infection can result in systemic inflammatory response syndrome with a high fatality rate, and there are currently no treatments or vaccines available. The microbiota has been implicated in host susceptibility to systemic viral infection and disease outcomes, but whether the gut microbiota is implicated in severe fever with thrombocytopenia syndrome virus (SFTSV) infection is unknown. Here, we analysed faecal and serum samples from patients with SFTS using 16S ribosomal RNA-sequencing and untargeted metabolomics, respectively. We found that the gut commensal *Akkermansia muciniphila* increased in relative abundance over the course of infection and was reduced in samples from deceased patients. Using germ-free or oral antibiotic-treated mice, we found that *A. muciniphila* produces the β -carboline alkaloid harmaline, which protects against SFTSV infection by suppressing NF- κ B-mediated systemic inflammation. Harmaline indirectly modulated the virus-induced inflammatory response by specifically enhancing bile acid-CoA: amino acid N-acyltransferase expression in hepatic cells to increase conjugated primary bile acids, glycochenodeoxycholic acid and taurochenodeoxycholic acid. These bile acids induced transmembrane G-protein coupled receptor-5-dependent anti-inflammatory responses. These results indicate the probiotic potential of *A. muciniphila* in mitigating SFTSV infection.

The dynamic cross-talk between the host and its microbiota bridged by microbial metabolites is not only essential for maintaining homeostasis^{1,2}, but is also important for regulating immune responses when the steady state is broken in the context of viral infection^{3,4}. The pre-existing

health status of the microbiota and alterations in it during the course of infection in an individual are likely to play an important (but still underestimated) role in determining susceptibility and resilience to infectious diseases.

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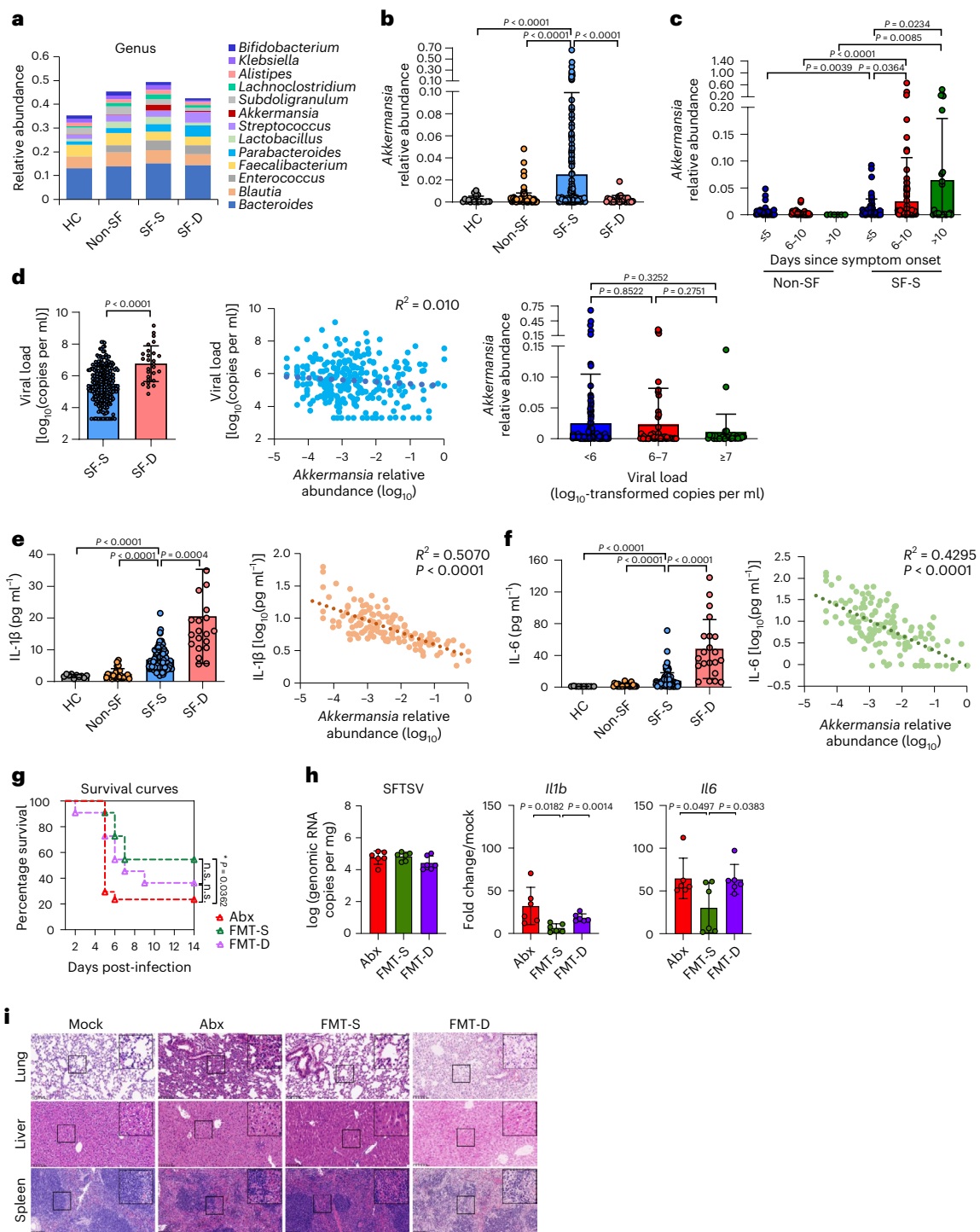


Fig. 1 | Surviving SFTSV-infected patients exhibited an increased faecal abundance of *Akkermansia* that was inversely correlated with the severity of systemic inflammatory responses. **a**, Relative abundance of genera among the four groups (HC $n = 19$, non-SF $n = 176$, SF-S $n = 233$, SF-D $n = 27$). **b**, Relative abundance of *Akkermansia* among the four groups (HC $n = 19$, non-SF $n = 176$, SF-S $n = 233$, SF-D $n = 27$). **c**, Relative abundance of *Akkermansia* during the three stages of illness between the SF-S and non-SF groups (non-SF $n = 176$, SF-S $n = 233$). **d**, Viral loads in the SF-S and SF-D groups and the correlation with *Akkermansia* relative abundance (SF-S $n = 233$, SF-D $n = 27$). **e**, **f**, Serum concentrations of IL-1 β (**e**) and IL-6 (**f**) among the four groups and the correlations with the relative abundance of *Akkermansia* (HC $n = 10$, non-SF $n = 25$, SF-S $n = 107$, SF-D $n = 21$). **g**, Survival kinetics analysis of Abx-treated and Abx-treated FMT recipient mice (mice that received FMT from recovered or

dead SFTSV-infected patients) that were pretreated with anti-IFNAR1 IgG and intraperitoneally inoculated with SFTSV (Abx $n = 16$, FMT-S $n = 11$, FMT-D $n = 11$). **h**, qPCR of SFTSV RNA (left), IL-1 β and IL-6 mRNA (right) in the spleens from Abx-treated mice with or without FMT (faeces from recovered or deceased SFTSV-infected patients) and infected with SFTSV at 3 d.p.i. ($n = 6$). **i**, HE staining of lung, liver or spleen cross-sections from Abx-treated mice with or without FMT (faeces from recovered SFTSV or deceased patients) and infected with SFTSV at 3 d.p.i. Boxed areas are magnified immediately in the top right corner. The two-sided P values were examined by Student's t test and data were presented as mean $\pm$ s.d. (**b–f**, **h**). R^2 and exact two-sided P values calculated by Pearson's test are shown (**d–f**). The Kaplan–Meier method and the log-rank test were used to analyse time-to-event data for treatment effect analysis (**g**).

The intestinal microbiome can inhibit viral infections locally and systemically through microbial metabolites or constituents^{5–8}. However, the molecular links between microbiota-derived metabolites and host immunity, as well as viral pathogenesis and disease outcomes, remain largely unknown, and a mechanistic understanding of the signalling pathways linking host inflammatory responses and microbial metabolites under systemic infection is still lacking.

Severe fever with thrombocytopenia syndrome (SFTS), which is an emerging tick-borne infectious disease, is caused by a novel *phlebovirus* belonging to the family Phenuiviridae, a segmented and negative-strand RNA virus that was originally discovered in mainland China in 2009^{9,10}. A total of 13,305 SFTS patients have been reported in 24 provinces (municipalities) in China up to December 2020¹¹, but the actual number of individuals infected with severe fever with thrombocytopenia syndrome virus (SFTSV) may be greatly underestimated¹². SFTSV infection has become pandemic in Asian countries^{13–17}. Despite the continuously expanding geographic distribution, no approved vaccines or specific antiviral treatments are currently available^{10,18}. Critically ill SFTS patients will develop a cytokine storm, leading to widespread tissue damage that results in multiple organ dysfunction syndrome with a high fatality rate of 12%–50%^{13,15,19}. Such features make SFTSV a perfect model for investigating whether and how the microbiota and specific microbial metabolites exert an effect on virus-induced systemic inflammatory response syndrome.

Here, we screened the intestinal microbiome of SFTS patients and identified that *Akkermansia muciniphila* was significantly more abundant in surviving patients than in deceased patients. Subsequently, we showed that *A. muciniphila* and its metabolite harmaline (HAL) can promote bile acid-CoA: amino acid *N*-acyltransferase (BAAT) expression in hepatocytes, which produces higher levels of glycochenodeoxycholic acid (GCDCA) and taurochenodeoxycholic acid (TCDCA) to reduce SFTS severity and mortality by suppressing the resulting systemic inflammatory responses via transmembrane G-protein coupled receptor-5 (TGR5)–NF- κ B signalling.

Results

Akkermansia muciniphila protects the host from SFTSV infection by attenuating systemic inflammation

Faecal samples were taken from 260 patients hospitalized with SFTS (surviving patients were designated as the SF-S group, and deceased patients were designated as the SF-D group; Tables S1 and S2), 176 non-SFTSV febrile patients (non-SF) and 19 healthy controls (HC) via 16S ribosomal RNA-sequencing (RNA-seq). We observed a significantly greater abundance of *Akkermansia*, *Lactobacillus*, *Enterococcus* and *Parabacteroides* in the SF-S group compared with the HC group (Fig. 1a), whereas only *Akkermansia* was markedly reduced in the SF-D group compared with the SF-S group (Fig. 1b and Extended Data Fig. 1a,b). Moreover, the abundance of *Akkermansia* increased with disease course in the SF-S group but remained constant in the non-SF group, implying that this phenotype was specific to SFTSV infection

(Fig. 1c). Systemic SFTSV burden was higher in the SF-D group than in the SF-S group but was not correlated with *Akkermansia* abundance in SFTS patients (Fig. 1d). By contrast, compared with SF-D patients, SF-S patients showed markedly lower expression of the proinflammatory cytokines interleukin-1 β (IL-1 β), interleukin-6 (IL-6) and tumour necrosis factor- α (TNF- α) in serum, the concentrations of which were inversely correlated with the relative abundance of *Akkermansia* (Fig. 1e,f and Extended Data Fig. 1c).

Subsequently, we prepared faecal samples pooled from four recovered SFTSV-infected patients (with a high abundance of *Akkermansia*) or from three patients who died of SFTSV infection (with a low abundance of *Akkermansia*) and performed faecal microbiota transplantation (FMT) in microbiota-depleted mice that were pretreated orally with a cocktail of four broad-spectrum antibiotics (hereafter referred to as the Abx mice)²⁰. As expected, FMT from SFTSV-infected patients into Abx-treated mice significantly restored the total faecal 16S rRNA copy number at day 0; however, it did not restore copy number to the same level as the PBS control (Extended Data Fig. 2a), thus implying less efficient colonization of human microbiota in the mouse gut. Moreover, the FMT experiments were not influenced by SFTSV faecal shedding because no viral genomic copy number was detectable in the stool of infected patients.

A lethal infection mouse model pretreated with an anti-interferon alpha receptor 1 (IFNAR1)-blocking antibody, was used to evaluate the pathogenesis of SFTSV infection throughout the entire study²¹. Remarkably, FMT from recovered patients (FMT-S group), but not from succumbed donors (FMT-D group), administered to Abx mice significantly increased the survival rate from 23% to 54% by the end of the observation period (Fig. 1g). Moreover, FMT-S mice exhibited greatly diminished proinflammatory cytokine expression in the spleens, and significantly improved pathological changes in the lung, liver and spleen, with similar splenic viral burden compared with those of non-FMT or FMT-D recipients (Fig. 1h,i).

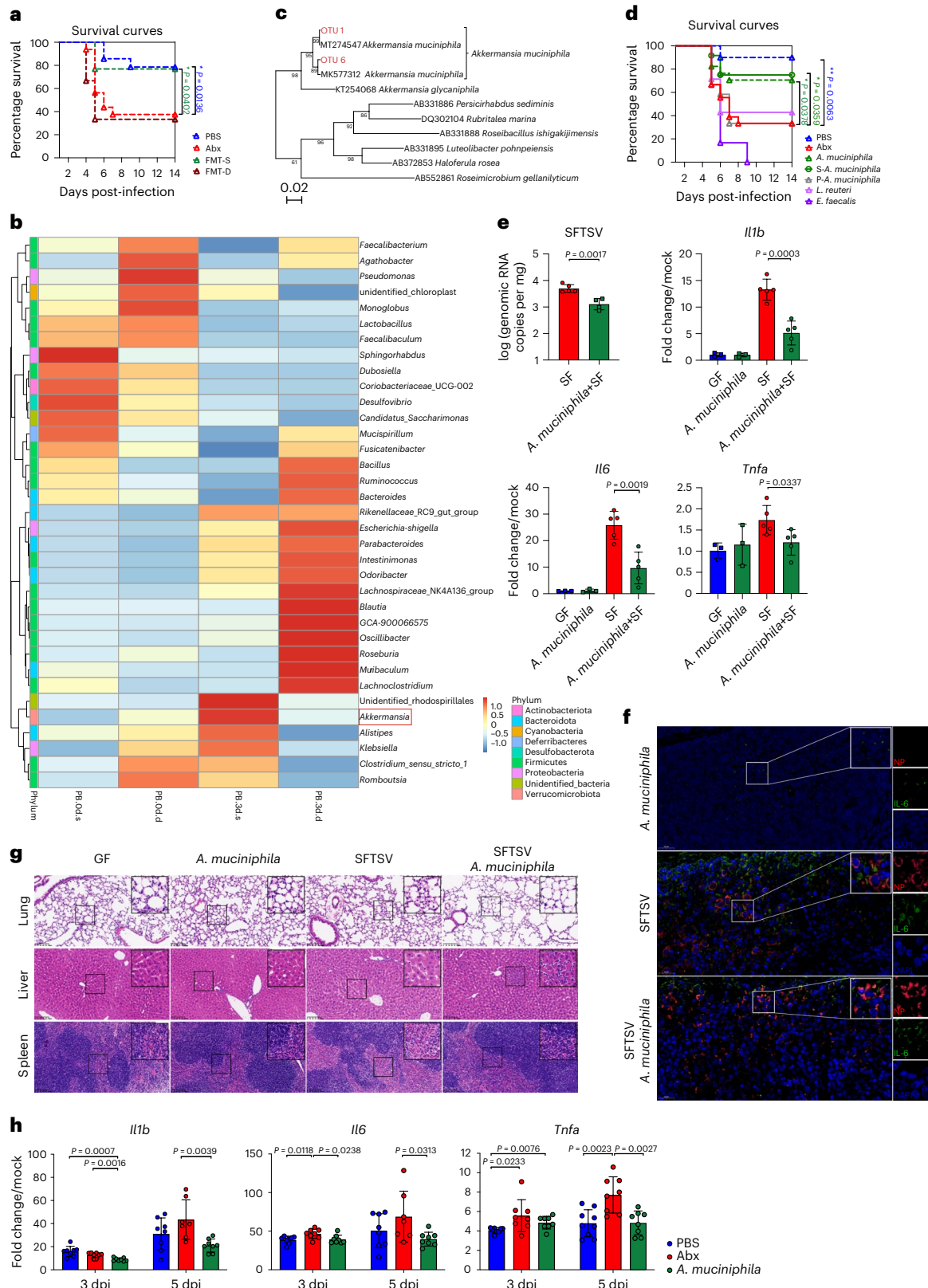
Because SFTSV infection caused over 60% mortality in Abx mice, but only approximately 20% lethality in PBS-treated mice, it implies that intestinal microbes confer protection against lethal SFTSV infection in our mouse model (Fig. 2a). Subsequently, we collected faecal samples from PBS-treated mice at 3 days post-infection (d.p.i.) with SFTSV and performed FMT experiments in Abx mice. Indeed, Abx mice that received FMT from surviving mice had a significantly enhanced survival rate of SFTSV infection, whereas FMT from donor mice that succumbed to infection exhibited no protective effect (Fig. 2a). Moreover, FMT-reconstituted Abx mice displayed significantly lower *Il1b* and *Il6* expression and ameliorated tissue damage in the lung, liver and spleen, but exhibited similar splenic viral burden to Abx mice without FMT (Extended Data Fig. 2b,c). Further faecal 16S rRNA analysis of PBS-treated mice at 3 d.p.i. revealed a significantly increased relative and absolute abundance of *Akkermansia* in the mice that survived, but not in those mice that succumbed to SFTSV systemic infection (Fig. 2b and Extended Data Fig. 2d).

Fig. 2 | Oral *A. muciniphila* administration decreases susceptibility to SFTSV infection and attenuates the resulting systemic inflammation in a mouse model. **a**, PBS-treated, Abx-treated and Abx-treated FMT recipient mice (mice that received FMT from recovered or dead SFTSV mice) were pretreated with anti-IFNAR1 IgG and inoculated intraperitoneally with SFTSV for survival kinetics analysis (PBS n = 14, Abx n = 16, FMT-S n = 13, FMT-D n = 6). **b**, Heatmap of the relative abundance of bacteria in faeces from mice infected with SFTSV at 0 and 3 d.p.i. **c**, Tree constructed based on 16S rRNA nucleotide sequences showing the phylogenetic relationships of two OTUs of *A. muciniphila* obtained in the study. **d**, Survival kinetics analysis of Abx-treated mice colonized with *A. muciniphila*, pasteurized *A. muciniphila* (P-*A. muciniphila*), supernatant *A. muciniphila* (S-*A. muciniphila*), *L. reuteri* or *E. faecalis* that were pretreated with anti-IFNAR1 IgG and inoculated intraperitoneally with SFTSV (PBS n = 10, Abx n = 18, *A. muciniphila* n = 17, P-*A. muciniphila* n = 12, S-*A. muciniphila* n = 12, *L. reuteri* n = 7, *E. faecalis* n = 6). **e**, Relative mRNA levels of splenic viral loads (left),

IL-1 β , IL-6 and TNF- α (right) normalized to mock animals of SFTSV-infected GF mice colonized with or without *A. muciniphila* at 3 d.p.i. (GF and *A. muciniphila* n = 3, SF and *A. muciniphila* plus SF n = 5). **f**, Representative IFA images of spleen sections from non-colonized or *A. muciniphila*-colonized GF mice infected with SFTSV at 3 d.p.i. SFTSV protein NP and IL-6 protein were double stained with the respective antibodies. **g**, HE staining of lung, liver or spleen cross-sections from SFTSV-infected GF mice colonized with or without *A. muciniphila* at 3 d.p.i. Boxed areas are magnified in the top right corner. Scale bar, 100 μ m. **h**, Relative mRNA expression of proinflammatory cytokines from spleen homogenates from Abx-treated mice colonized with or without *A. muciniphila* and infected with SFTSV at 3 and 5 d.p.i. (n = 8). The two-sided *P* values were examined by Student's *t* test and data were presented as mean $\pm$ s.d. (**e,h**). The Kaplan–Meier method and the log-rank test were used to analyse time-to-event data for treatment effect analysis (**a,d**).

We further selected 20 representative faecal samples from recovered patients with high *Akkermansia* abundance for deep sequencing and identified two operational taxonomic units (OTUs) belonging to *A. muciniphila*, with OTU1 detected in 15 samples and OTU6 in 5 samples (Fig. 2c). To determine the role of *A. muciniphila* in the protection against SFTSV infection, we used live and pasteurized *A. muciniphila*

to gavage Abx mice and then inoculated them with SFTSV. Because the relative abundance of *Lactobacillus* was also increased, whereas the relative abundance of *Enterococcus* was decreased in the surviving SFTSV-infected patients (Fig. 1a), a well-studied human symbiotic isolate *Lactobacillus reuteri* (which belongs to the *Lactobacillus* genus) and *Enterococcus faecalis* (which is a major cause of nosocomial infection)



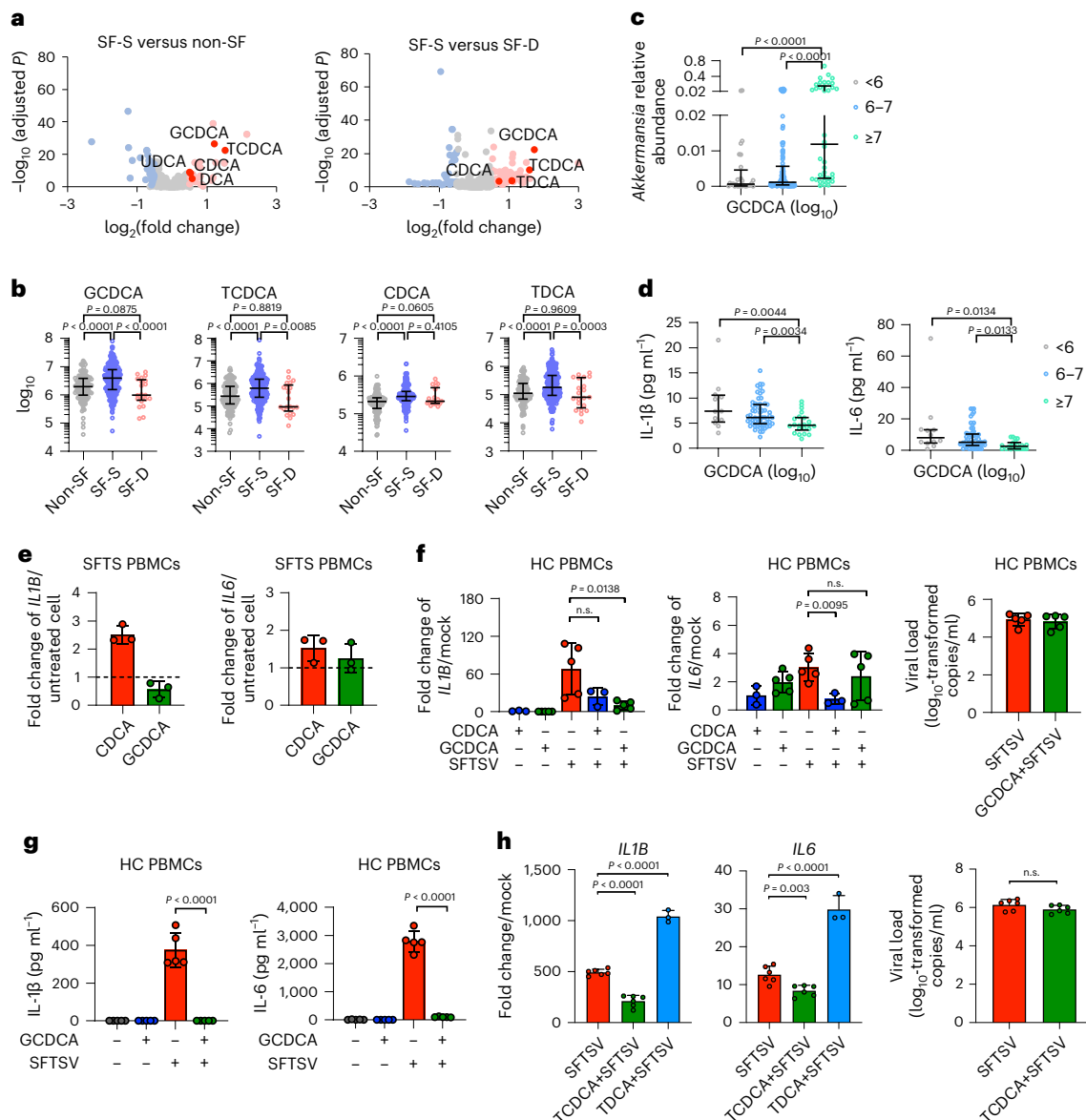


Fig. 3 | Surviving SFTSV-infected patients displayed an *A. muciniphila*-related increase in serum GCDCA and TCDCA that suppressed inflammatory responses in human PBMCs. **a**, Volcano plot showing differentially expressed metabolites between the two groups (left, SF-S versus non-SF; right, SF-S versus SF-D). Each dot indicates an individual metabolite, coloured red when a metabolite is significantly upregulated and blue when a gene is significantly downregulated in the SF-S group. **b**, Comparison of GCDCA, TCDCA, CDCA and TDCA levels among the three groups (non-SF $n = 132$, SF-S $n = 222$, SF-D $n = 21$). **c**, Relative abundance of *Akkermansia* (<6 $n = 28$, 6–7 $n = 131$, ≥7 $n = 38$). **d**, Serum concentrations of IL-1β and IL-6 are associated with the levels of GCDCA (<6 $n = 12$, 6–7 $n = 61$, ≥7 $n = 22$). **e**, Relative mRNA levels of IL-1β and IL-6 in SFTS

patient PBMCs treated with 100 μM GCDCA or 100 μM CDCA for 24 h ($n = 3$). **f**, Relative mRNA levels of IL-1β, IL-6 (left) and SFTSV RNA (right) in healthy control PBMCs pretreated with 100 μM GCDCA or 100 μM CDCA and then infected with SFTSV (MOI 1) at 24 h post-infection ($n = 5$). **g**, IL-1β and IL-6 protein expression in healthy control PBMCs pretreated with 100 μM GCDCA and then infected with SFTSV (MOI 1) at 24 h post-infection was determined by ELISA ($n = 5$). **h**, Relative mRNA levels of IL-1β, IL-6 (left) and SFTSV RNA (right) in healthy control PBMCs pretreated with 100 μM TCDCA or 100 μM TDCA and then infected with SFTSV (MOI 1) at 24 h post-infection (SFTSV and TCDCA plus SFTSV $n = 6$, TDCA plus SFTSV $n = 3$). The two-sided P values were examined by Student's t test and data were presented as mean ± s.d.

were used as controls of unrelated commensal bacteria. Strikingly, *A. muciniphila* colonization significantly protected Abx mice from lethal SFTSV infection, whereas *L. reuteri* gavage did not alter the lethality rate, and *E. faecalis* even exacerbated the mortality rate (Fig. 2d), despite effective colonization (Extended Data Fig. 2e).

To further confirm the role of *A. muciniphila* in modulating host inflammatory responses against SFTSV infection, we performed bacteria monocolonization in a gnotobiotic mouse model. At 3 d.p.i., significantly greater titres of SFTSV were detected in the spleen of non-colonized germ-free (GF) mice compared with in

A. muciniphila-colonized GF mice (Fig. 2e). Moreover, non-colonized GF mice exhibited more robust proinflammatory cytokine expression, as well as more severe tissue inflammatory lesions, compared with GF mice colonized by *A. muciniphila* (Fig. 2e–g). Similarly, in Abx mice, *A. muciniphila* reconstitution reversed SFTSV titre differences in the spleen, liver and lung at 3 d.p.i. (Extended Data Fig. 2f) and decreased the expression of *Il1b*, *Il6* and *Tnfa* at the transcriptional and protein levels (Fig. 2h and Extended Data Fig. 2g). At 5 d.p.i., SFTSV titres were higher in the spleen, but not in the liver or lung of Abx compared with PBS-treated and *A. muciniphila*-colonized Abx mice (Extended Data

Fig. 2f). Even without an obvious reduction in viral titres, *A. muciniphila* colonization significantly diminished proinflammatory cytokine expression (Fig. 2h) and alleviated tissue damage and inflammatory infiltration in tissues at 5 d.p.i. (Extended Data Fig. 2h).

To define the protective component of *A. muciniphila*, wild-type (WT) B6 mice were gavaged with filtered *A. muciniphila* supernatant or pasteurized *A. muciniphila*, which was initiated in parallel with Abx administration until the end of the experiments (Extended Data Fig. 1d), to maintain the level of active metabolites or pasteurized *A. muciniphila* cells. Interestingly, the administration of Abx-treated mice with *A. muciniphila* supernatant significantly alleviated SFTSV-induced mortality, whereas gavage with pasteurized *A. muciniphila* did not alter mortality (Fig. 2d), even though the pasteurized cells were precisely detected in the faeces of treated animals before SFTSV infection (Extended Data Fig. 2e), suggesting that the protective effect was very likely dependent on the microbial metabolites rather than the bacterial components. Together, these concordant results in both GF and Abx animals indicate a role for *A. muciniphila*-driven metabolites in alleviating inflammatory responses in both the peripheral and distal organs after SFTSV systemic infection.

***A. muciniphila*-driven conjugated primary bile acid protects the host from SFTSV infection by dampening systemic inflammatory responses**

We next performed untargeted metabolomics analysis on a total of 405 serum samples collected from SF-S ($n = 222$), SF-D ($n = 21$) and non-SF ($n = 132$) patients, and detected 69 metabolites that were differentially regulated between the SF-S and non-SF groups and identified 153 differential metabolites between the SF-S and SF-D groups (Fig. 3a). Among these differential metabolites, a series of bile acids (BAs), including chenodeoxycholic acid (CDCA), GCDCA, TCDCA and taurodeoxycholic acid (TDCA), were found to be significantly increased in the SF-S group compared with the other two groups (Fig. 3a,b). Furthermore, a high GCDCA serum concentration was positively correlated with an increased relative abundance of *A. muciniphila* and negatively correlated with IL-1 β and IL-6 expression in serum (Fig. 3c,d).

CDCA is predominantly conjugated to glycine (to form GCDCA) and rarely conjugated to taurine (which would form TCDCA) in humans²². Peripheral blood mononuclear cells (PBMCs) from three SFTS patients in the acute phase treated with GCDCA exhibited significantly reduced expression of *IL1B* (but not *IL6*), compared with the untreated control (dashed line), whereas CDCA increased *IL1B* expression by 2.5-fold (Fig. 3e and Extended Data Fig. 3a). In addition, we pretreated PBMC extracts from five healthy donors with CDCA or GCDCA and discovered a remarkable decrease in IL-1 β and IL-6 expression and comparable viral replication in the GCDCA-pretreated PBMCs compared with the untreated control post SFTSV infection (Fig. 3f,g and Extended Data Fig. 3b). TCDCA pretreatment also resulted in significantly decreased *IL1B* and *IL6* transcripts independent of viral replication, whereas TDCA pretreatment greatly increased proinflammatory cytokine expression (Fig. 3h).

Next, we infected PBS-, Abx- or *A. muciniphila*-colonized Abx mice with SFTSV and subjected the serum samples to untargeted

metabolomics analyses. Partial least-squares discrimination analysis revealed significantly different metabolomics profiles of infected mice compared with mock mice (Extended Data Fig. 4a). KEGG analyses indicated that the differential metabolites in PBS- or *A. muciniphila*-colonized mice compared with Abx mice were indeed enriched in bile secretion and cholesterol metabolism (Fig. 4a and Extended Data Fig. 4b). Furthermore, the relative abundances of GCDCA and a series of taurine-conjugated BAs, including TCDCA, taurocholate acid (TCA), TDCA and taurocholate- α -muricholic acid (T- α -MCA), were markedly elevated in the serum of mice colonized with *A. muciniphila* (Fig. 4b,c). Among them, GCDCA and TCDCA significantly downregulated *Il6*, *Il1b* and *Tnfa* expression in SFTSV-infected mouse PBMCs, whereas TCA, TDCA and T- α -MCA had no or even an augmenting effect on proinflammatory cytokine expression compared with unprimed cells independent of viral replication (Fig. 4d and Extended Data Fig. 5a,b). In addition, we confirmed that TCDCA downregulated proinflammatory cytokine expression in a dose-dependent, viral replication-independent manner (Extended Data Fig. 5c,d).

Because the vast majority of CDCA is conjugated to taurine instead of glycine in rodents²², we administered Abx mice with TCDCA in drinking water for 4 weeks to test whether TCDCA is the proximate BA metabolite that could likewise ameliorate inflammatory responses post SFTSV infection. Interestingly, TCDCA treatment significantly increased its serum concentration (Extended Data Fig. 5e) and reduced the mortality rate of infected Abx mice by 40% (Fig. 4e). Moreover, we observed significantly reduced proinflammatory cytokine expression and alleviated tissue damage, but no altered viral replication in TCDCA-treated mice compared with Abx control mice (Fig. 4f–h and Extended Data Fig. 5f). These findings suggest a role for *A. muciniphila* in the induction of inflammation-suppressing GCDCA and TCDCA, which can dampen host inflammatory damage resulting from SFTSV infection in vivo.

The *A. muciniphila* metabolite HAL suppresses systemic inflammatory responses resulting from SFTSV infection by upregulating BAAT expression in hepatocytes

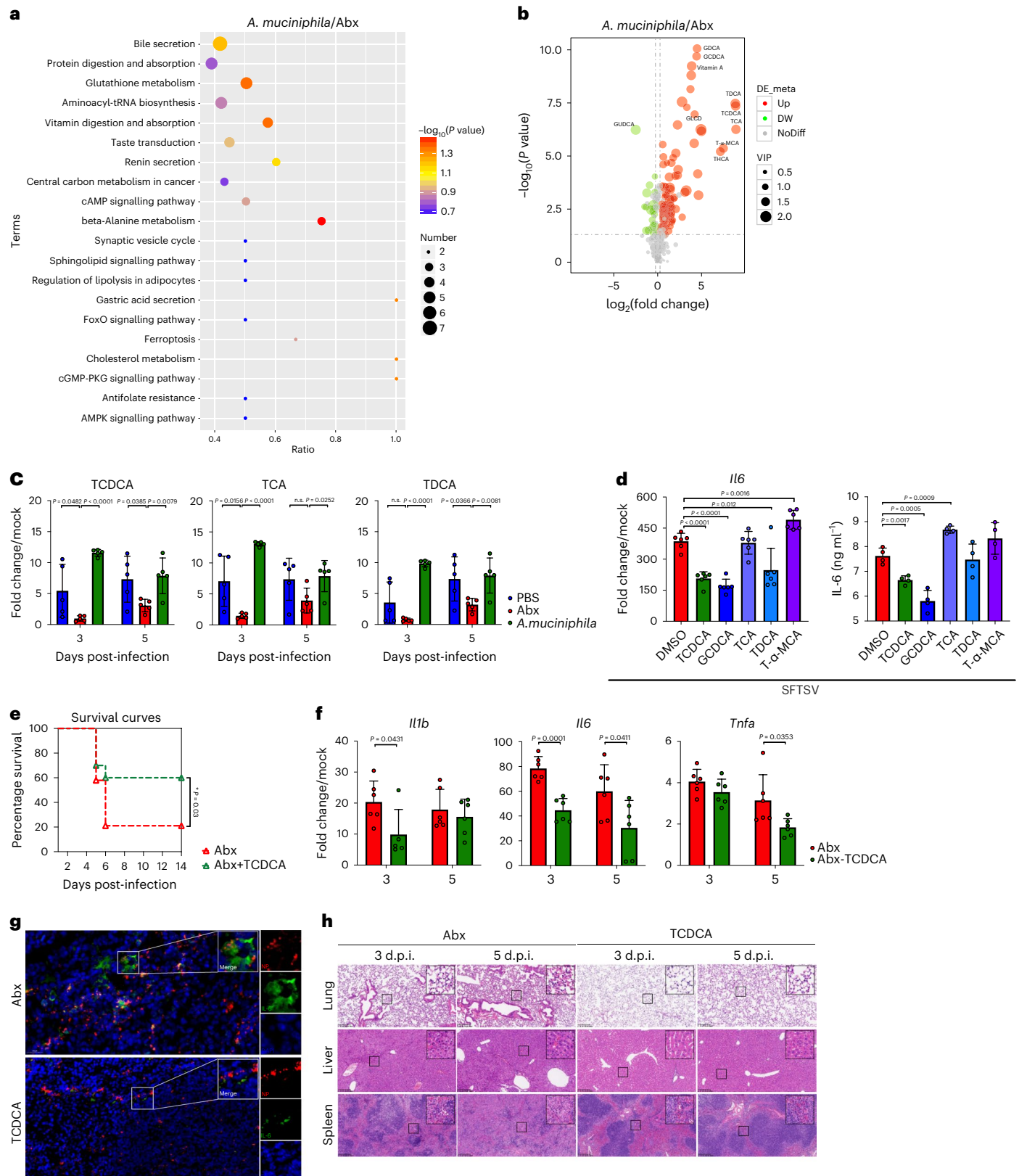
Hepatocytes synthesize primary BAs via hydroxylation of cholesterol to generate cholic acid and CDCA, and subsequently conjugate to either glycine or taurine by bile acyl-CoA synthetase (BACS) and BAAT²³. First, we excluded the possibility that *A. muciniphila* colonization alone produces more cholesterol to generate more conjugated primary BAs by showing comparable hepatic concentrations of cholesterol among PBS-, Abx- and *A. muciniphila*-colonized mice (Extended Data Fig. 6a). We also discovered an equivalent upregulation of the apical sodium-dependent BA transporter in the ileum of the Abx- or *A. muciniphila*-colonized mice compared with PBS controls (Extended Data Fig. 6b), suggesting that the reabsorption of conjugated primary BAs into the circulation does not account for the *A. muciniphila*-associated protection. Regarding the impact of *A. muciniphila* colonization on TCDCA/GCDCA biosynthesis, we determined that CYP7A1, CYP8B1, CYP27A1, CYP7B1 and BACS expression was not significantly altered, whereas the expression of BAAT was markedly augmented (Fig. 5a and Extended Data Fig. 6c,d).

Fig. 4 | *A. muciniphila*-driven conjugated primary BA TCDCA protects host from SFTSV infection by dampening systemic inflammatory responses in mouse model. **a**, KEGG analysis of differentially regulated metabolites between the *A. muciniphila* group and the Abx group. **b**, Volcano plot of differentially regulated metabolites between the *A. muciniphila* group and the Abx group. **c**, Fold change of taurine-conjugated BA abundance in serum collected from PBS-treated, Abx-treated or *A. muciniphila*-colonized mice infected with SFTSV at 3 and 5 d.p.i. over the mock group ($n = 5$). **d**, mRNA transcripts (left, $n = 6$) or protein levels of IL-6 (right, $n = 4$) in SFTSV-infected mouse PBMCs pretreated with TCDCA, GCDCA, TCA, TDCA or T- α -MCA (MOI1) at 24 h post-infection. **e**, Groups of Abx mice pretreated with vehicle or TCDCA were administered anti-IFNAR1 IgG and subsequently intraperitoneally inoculated with SFTSV for

survival kinetics analysis (Abx $n = 19$, Abx + TCDCA $n = 20$). **f**, Relative expression levels of IL-1 β , IL-6 and TNF- α in spleens from Abx-treated mice treated with or without TCDCA and infected with SFTSV at 3 and 5 d.p.i. ($n = 6$). **g**, IFA images of spleen sections from Abx-treated mice treated with or without TCDCA at 3 d.p.i. SFTSV protein NP and IL-6 protein were double stained with the respective antibodies. **h**, HE staining of lung, liver or spleen cross-sections from Abx-treated mice treated with or without TCDCA and infected with SFTSV at 3 and 5 d.p.i. Boxed areas are magnified immediately in the top right corner. The two-sided P values were examined by Student's t test and data were presented as mean $\pm$ s.d. (**a–d,f**). The Kaplan–Meier method and the log-rank test were used to analyse time-to-event data for treatment effect analysis (**e**).

Because *L. reuteri* colonization did not protect Abx mice from SFTSV infection (Fig. 2d), we used *L. reuteri* as an unrelated control and demonstrated that no upregulated levels of BAAT were detected in the livers of *L. reuteri*-colonized mice in comparison with PBS or Abx mice (Fig. 5a and Extended Data Fig. 6d). Analogous to the Abx model, GF mice colonized with *A. muciniphila* also had markedly higher BAAT

levels in hepatic tissues compared with vehicle-treated GF mice (Fig. 5b and Extended Data Fig. 6e,f). As expected, *A. muciniphila*-colonized Abx mice or GF mice had remarkably augmented serum levels of TCDCA, as well as other taurine-conjugated BAs, including T- α -MCA and TCA, compared with uncolonized Abx or GF mice (Fig. 5c and Extended Data Fig. 6g).



Because primary BAs are produced by hepatocytes, and modified and bio-transformed by commensal microbiota²⁴, we subsequently hypothesized that *A. muciniphila* generates certain bioactive small metabolites to enhance BAAT-driven TCDCA production and to ultimately yield protection from SFTSV infection. Therefore, we fractionated the *A. muciniphila* cultured supernatants to generate a series of fractions based on molecular mass²⁵. Intriguingly, the 10 kDa or less filtrate substantially augmented BAAT expression in a dose-dependent manner in the Huh-7 cell line (similar to what occurred in the unfractionated supernatant control), whereas this phenotype was not observed for the other filtrates (Extended Data Fig. 6h,i). Treatment with proteinase K did not interfere with the effect of the 10 kDa filtrate, thus indicating that the effective factors responsible for the induction of BAAT expression are very likely small metabolic molecules or small microbial proteins that are not affected by proteinase K (Extended Data Fig. 6i).

To investigate the bacterial metabolites mediating BAAT augmentation, we examined the supernatant of *A. muciniphila* cultured in brain heart infusion medium via untargeted metabolome analyses. Major metabolites that *A. muciniphila* produces (such as acetate and propionate) were identified using gas chromatography mass spectrometry (Extended Data Fig. 6j). In addition, liquid chromatography–tandem mass spectrometry (LC–MS/MS) analysis yielded five *A. muciniphila*-associated metabolites with relatively higher concentrations in both positive and negative ion-exchange chromatography (Extended Data Fig. 7a). The concentration of HAL was positively correlated with *A. muciniphila* cell number (Extended Data Fig. 7b). We pretreated Huh-7 cells with the five metabolites, and one of them, HAL, appeared to induce markedly higher levels of BAAT in both the Huh-7 cell line and mouse primary hepatocytes (MPHs) in a dose-dependent manner (Fig. 5d,e and Extended Data Fig. 7c,d). Because *A. muciniphila* releases short-chain fatty acids in vivo and in vitro¹, we included acetate in the study as an unrelated metabolite control and observed no effect on BAAT expression (Fig. 5e). Furthermore, we administered HAL together with CDCA to MPHs and found that the concentration of CDCA decreased significantly, whereas the concentration of TCDCA increased markedly. This phenotype was lost when small interfering RNAs targeting BAAT were transfected into MPHs, implying that the taurine-conjugating metabolic activities were driven by HAL in a BAAT-dependent manner (Fig. 5f and Extended Data Fig. 7e). Consistently, HAL markedly increased expression of BAAT but not CYP7A1, CYP8B1, CYP27A1, CYP7B1 or BACS in hepatic tissues of Abx animals, in comparison with expression in untreated Abx mice (Fig. 5g and Extended Data Fig. 7f,g), whereas acetate induced significantly lower levels of BAAT compared with HAL (Fig. 5g). As expected, HAL-treated Abx mice or GF mice had markedly augmented serum TCDCA levels compared with uncolonized Abx or GF mice (Extended Data Fig. 8a,b). In addition, Abx mice had no HAL in their serum samples, whereas Abx mice colonized with *A. muciniphila* had picomolar quantities of HAL in serum (Extended Data Fig. 8c). Notably, a higher concentration of

HAL was determined in serum from recovered SFTSV-infected patients compared with recovered febrile patients without SFTSV infection (Extended Data Fig. 8d). These concordant results demonstrated that the *A. muciniphila*-derived metabolite HAL promotes BAAT expression both in vitro and in vivo.

A further in vivo protective experiment showed that HAL-treated Abx mice exhibited an increased survival rate compared with non-HAL-treated, acetate-treated or propionate-treated Abx mice, albeit to a lesser extent than PBS-treated controls (Fig. 5h), thus suggesting an anti-SFTSV effect exerted by HAL in the context of microbiota deficiency. Systemic inflammatory cytokine levels as well as SFTSV-associated histopathological changes in various tissues were apparently ameliorated in HAL-treated GF mice in comparison with vehicle-treated controls (Fig. 5i, Extended Data Fig. 8e,f). We believe the inflammation-suppressing effect was not directly exerted by HAL because mouse PBMCs pretreated with HAL displayed equivalent pro-inflammatory cytokine expression compared with the vehicle control post SFTSV infection (Extended Data Fig. 8g). Moreover, HAL treatment failed to provide efficient protection for Abx WT B6 mice transiently transfected with *siBAAT* (Fig. 5j and Extended Data Fig. 8h), suggesting that HAL confers protection in a BAAT-dependent manner.

Conjugated primary BA GCDCA suppresses NF-κB-mediated inflammatory responses in a manner dependent on the BA receptor TGR5

To investigate the molecular mechanisms by which GCDCA suppresses SFTSV-induced inflammation, we used a human cellular model, THP-1 cells, because monocytes are recognized as the main target cells for SFTSV infection in human PBMCs^{21,26,27}. Indeed, GCDCA drastically downregulated IL-1β and IL-6 in SFTSV-infected THP-1 cells in a dose-dependent and viral replication-independent manner (Fig. 6a–c and Extended Data Fig. 9a,b), whereas CDCA significantly upregulated *IL1B* and *IL6* (Extended Data Fig. 9c,d).

Furthermore, we employed transcriptome analysis of SFTSV-infected THP-1 cells with or without GCDCA pretreatment and found that a large pool of inflammatory response-related genes was specifically upregulated in SFTSV-infected versus mock THP-1 cells, according to Gene Ontology (GO) analysis, whereas significant downregulation of TLR8 and downstream signalling pathways was observed in GCDCA-pretreated infected cells (Fig. 6d,e). However, the knock-down of *TLR8* and *MyD88*, GCDCA still inhibited the expression of IL-1β and IL-6 under SFTSV infection (Extended Data Fig. 10a,b).

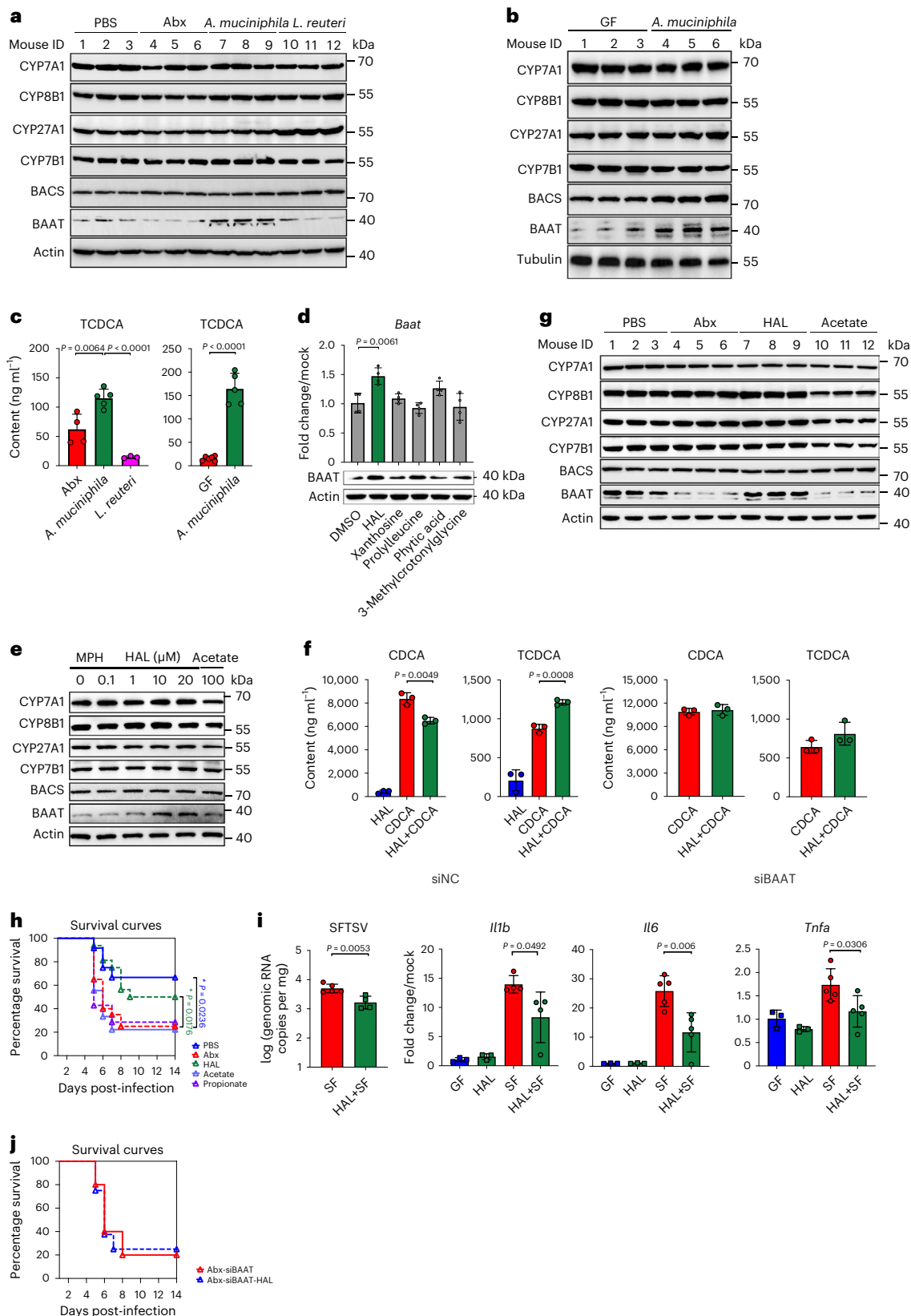
In total, several genes in the NF-κB signalling pathway were significantly downregulated in GCDCA-pretreated, SFTSV-infected THP-1 cells, including *RELA* and *NFKB1* (Fig. 6e). We further demonstrated that GCDCA pretreatment markedly decreased the level of the P50 subunit and the phosphorylated form of the P65 subunit (p-P65) in both the cytosolic and nuclear fractions, as well as specifically reducing the P65 level in the nucleus in a MyD88 signalling-independent manner (Fig. 6f and Extended Data Fig. 10c,d).

Fig. 5 | *A. muciniphila* metabolite HAL enhances TCDCA production by upregulating the expression of BAAT in a mouse model. a,b, Enzymes of BA biosynthesis pathways in the livers of (a) PBS-treated, Abx-treated, *A. muciniphila*-colonized or *L. reuteri*-colonized mice and (b) non-colonized or *A. muciniphila*-colonized GF mice at 5 days post-colonization. **c**, Serum TCDCA concentration of Abx-treated, *A. muciniphila*-colonized or *L. reuteri*-colonized mice (left) and GF mice colonized with or without *A. muciniphila* (right) at 5 days post-colonization (Abx *n* = 4, *A. muciniphila* *n* = 5, *L. reuteri* *n* = 3, GF *n* = 6). **d**, Relative mRNA (upper) and protein (lower) expression of Huh-7 cells treated with *A. muciniphila*-associated metabolites at 24 h post-treatment (*n* = 4). **e**, Enzymes of BA biosynthesis pathways in MPHs treated with different doses of HAL or 100 μM sodium acetate at 24 h post-treatment. **f**, Concentration of CDCA or TCDCA in culture supernatants of MPHs transfected with Negative control small interfering RNA or BAAT small interfering RNA (*siBAAT*) and treated with 50 μM CDCA plus 10 μM HAL at 24 h post-treatment (*n* = 3). **g**, Western blot of BA

biosynthesis enzymes in the livers of PBS-, Abx-, HAL- or acetate-treated mice at 5 days post-treatment. **h**, Groups of PBS-treated, Abx-treated and Abx mice treated with HAL, acetate or propionate were administered anti-IFNAR1 IgG and inoculated intraperitoneally with SFTSV for survival kinetics analysis (PBS *n* = 12, Abx *n* = 17, HAL *n* = 16, acetate *n* = 9, propionate *n* = 7). **i**, qPCR of SFTSV RNA (left) and IL-1β, IL-6 or TNF-α mRNA (right) in spleens from GF mice treated with or without HAL and infected with SFTSV at 3 d.p.i. (GF and HAL *n* = 3, SF and SF plus HAL *n* = 5). **j**, Groups of Abx-treated WT B6 mice transiently transfected with *siBAAT* and treated with dimethylsulfoxide or HAL were administered anti-IFNAR1 IgG and intraperitoneally inoculated with SFTSV for survival kinetics analysis (Abx-*siBAAT* *n* = 10, Abx-*siBAAT*-HAL *n* = 8). The two-sided *P* values were examined by Student's *t* test and data were presented as mean ± s.d. (**c,d,f,i**). The Kaplan–Meier method and the log-rank test were used to analyse time-to-event data for treatment effect analysis (**h,j**).

BAs regulate innate immune responses by activating different receptors, particularly TGR5 and farnesoid X receptor (FXR). TGR5, as a BA-activated membrane receptor, exhibits anti-inflammatory function in macrophages. The strongest activators of TGR5 are lithocholic acid and deoxycholic acid. CDCA has the greatest FXR-activating potential, followed by deoxycholic acid, lithocholic acid and finally cholic

acid^{28–30}. To identify the receptor(s) that GCDCA utilizes to confer anti-inflammatory responses in the context of SFTSV infection, we knocked down TGR5 and FXR using siRNA followed by GCDCA treatment and SFTSV infection (Fig. 6h and Extended Data Fig. 10h). Intriguingly, the suppressive effect of GCDCA pretreatment on SFTSV-induced IL-1 β and IL-6 upregulation was significantly impaired under *TGR5*



(but not *FXR*) knockdown (Fig. 6g and Extended Data Fig. 10g,h). Furthermore, following *TGR5* depletion, the differences in p50 and p-P65 protein levels were also lost in GCDCA-treated cells compared with the non-treated controls (Fig. 6h and Extended Data Fig. 10e,f). Correspondingly, the significant difference in the survival rate between the TCDCa-treated and untreated Abx WT mice (Fig. 4e) was completely lost in the TCDCa-treated and untreated Abx *TGR5*^{-/-} mice (Fig. 6i and Extended Data Fig. 10i). Consistent with these results, the protective effect of *A. muciniphila* colonization (Extended Data Fig. 10j) also required *TGR5* signalling, because the fatality rate was not restored in *TGR5*^{-/-} mice after *A. muciniphila* reconstitution (Fig. 6j). Together, these data indicate that *A. muciniphila*-driven TCDCa confers protection against SFTSV systemic infection via *TGR5* signalling in vivo.

Discussion

In this study, we described an *A. muciniphila*–BA–*TGR5* axis that limits host NF- κ B-mediated immunopathogenic responses resulting from infections of SFTSV and potentially other systemic viral pathogens (Extended Data Fig. 10k). Critically ill and deceased patients with SFTS showed a significantly lower abundance of faecal *A. muciniphila*, thus suggesting a potential role for *A. muciniphila* as a microbial biomarker in predicting the outcome of systemic SFTSV infection (Fig. 1b). One limitation of this study was that we could not acquire faecal samples from patients before they developed clinical symptoms or before they were hospitalized and diagnosed, which represents a major challenge for establishing the clinical connectivity between the pre-existing status of *A. muciniphila* and SFTSV infection. Nevertheless, in our mouse model, we observed significantly increased *A. muciniphila* abundance at 3 d.p.i. in SFTSV-infected mice that survived compared with those that succumbed to infection. By contrast there was no difference in *A. muciniphila* relative abundance before SFTSV infection between animals that survived and those that succumbed (Fig. 2b), thus indicating that survival may not be due to the pre-existence of intestinal *A. muciniphila*, whereas SFTSV infection could somehow increase the abundance of this specific commensal bacterium. Another example of viral infection-mediated alterations on microbiome composition was reported by Deriu et al., who demonstrated that type I interferons induced by influenza virus promote the depletion of obligate anaerobic bacteria and the enrichment of *Proteobacteria* in the gut³¹.

The effect of *A. muciniphila* on diminishing systemic inflammatory responses has been mentioned recently in the literature, with studies reporting that *A. muciniphila* exerts an anti-influenza effect by lowering pulmonary viral titres, reducing proinflammatory cytokine expression and enhancing the levels of type I and type II interferons in mice that were orally administered live or pasteurized *A. muciniphila*³². By contrast, we have previously demonstrated that *A. muciniphila* monocolonization in Abx mice did not restore the lethality upon encephalomyocarditis virus systemic infection²⁰, which is suggestive of a virus-specific manner of protection mediated by *A. muciniphila*.

A. muciniphila colonization can increase a series of BAs upon SFTSV infection (Fig. 4a,b), which improves current understanding of the role of *A. muciniphila* in generating acetate, propionate, succinate, ethanol and sulfate during mucin fermentation^{33,34}. The correlation between an increased abundance of caecal *A. muciniphila* and

higher levels of circulating primary BAs had previously been reported only in a mouse model that underwent bile diversion surgery³⁵. The current study demonstrated that the *A. muciniphila* metabolite HAL can regulate primary BA conjugation by specifically enhancing BAAT expression in hepatocytes (Fig. 5a). These data are partially consistent with research conducted by Sayin et al., who reported that the gut microbiota regulates expression of CYP7A1, CYP7B1, CYP27A1 and BACS, but not of CYP8B1 or BAAT³⁶. We speculate that the inconsistency between the two studies might be attributed to the different microbiota-depletion mouse models and distinct testing techniques used, because we compared the expression of enzymes between Abx and conventional mice, not between GF and conventional mice, and used the less-sensitive western blotting method instead of quantitative polymerase chain reaction (qPCR).

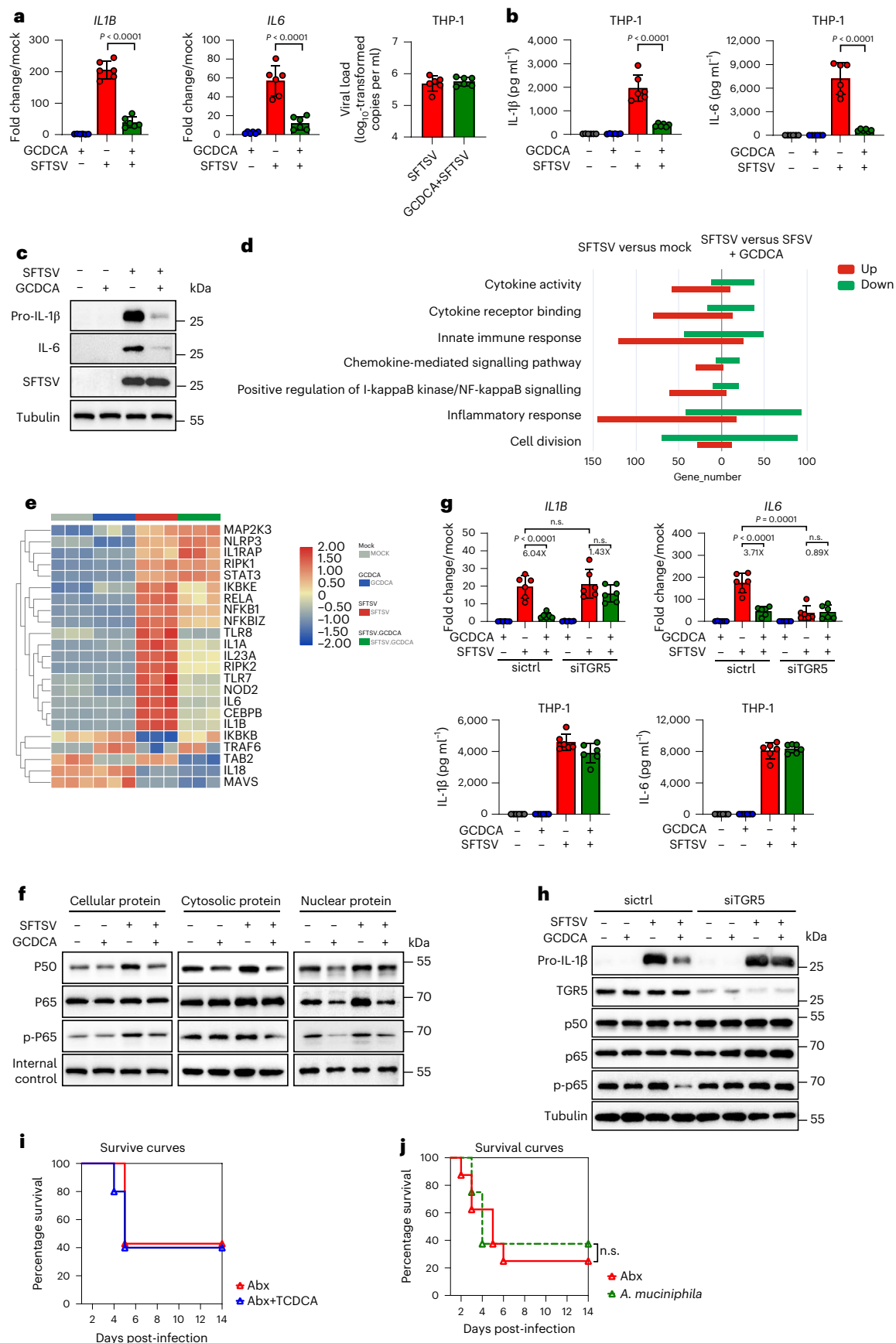
As a fluorescent psychoactive compound that was first isolated from the seeds of *P. harmala* in 1841, the ancient metabolite HAL has been extensively studied for its therapeutic potential and effectiveness in vasorelaxant, hypothermic, antimicrobial and other pharmacological activities³⁷. However, it remains too early to discuss its clinical utilization for treating SFTSV systemic infection because elevated dosages of HAL can cause agitation, cytotoxicity, delirium, paralysis or visual issues^{38,39}. Further work is needed to confirm whether other intestinal bacteria can produce HAL. Future studies will be performed to identify which gene(s) is/are involved in HAL synthesis within the genome of *A. muciniphila*. Moreover, establishment of a mutant *A. muciniphila* strain with the specific silenced gene can help us to build direct correlations between *A. muciniphila* and HAL, and would enable mechanistic dissections of the specific role of *A. muciniphila*-synthesized HAL for mitigating SFTSV systemic infections. Chen et al. identified a new microbial Pictet–Spenglerase NscbB from *Nocardiopsis synnemataformans* DSM 44143 that is involved in the production of β -carboline alkaloid⁴⁰. We analysed the entire genome of *A. muciniphila* through homologous alignment with the sequences of NscbB but failed to identify any counterpart genes that may encode the enzyme catalysing construction of the β C skeleton. Therefore, the specific genes of *A. muciniphila* involved in HAL synthesis still need to be further studied.

Poles et al. demonstrated that macrophages treated with the *TGR5* agonist INT-777 displayed higher cellular concentrations of cyclic AMP (cAMP), and activated protein kinase A and cAMP-responsive element binding protein, which are believed to inhibit STAT1 phosphorylation and NF- κ B transcriptional activity^{41,42}. However, we observed no variation in the cAMP–protein kinase A–cAMP-responsive element binding protein pathway in our transcriptome analysis of SFTSV-infected groups with or without GCDCA pretreatment (Fig. 6e), indicating that GCDCA may interfere with SFTSV-induced NF- κ B activation via other signalling pathways. Therefore, future investigations are needed to determine the missing molecular link between BA signalling and NF- κ B suppression under SFTSV infection.

Our findings build on the very limited understanding of the signalling pathways linking host anti-inflammatory responses and microbial metabolites under systemic viral infection. More importantly, it may have implications for the rational design of microbiome-based diagnostics and therapeutics for individuals at risk of critical systemic infection.

Fig. 6 | GCDCA suppresses NF- κ B-mediated inflammatory responses via *TGR5* signalling. a,b, THP-1^{PMA} cells were pretreated with 100 μ M GCDCA and then infected with SFTSV (MOI 1) for 24 h. **a**, Relative mRNA levels of IL-1 β , IL-6 (left) and SFTSV RNA (right) in THP-1^{PMA} cells ($n = 6$). **b**, IL-1 β and IL-6 protein expression in THP-1^{PMA} cells was determined by using ELISA ($n = 6$). **c**, Western blot of pro-IL-1 β , IL-6, SFTSV NP and tubulin in THP-1^{PMA} cells. **d**, Transcriptomic ontology (GO) analysis of differentially regulated gene numbers. **e**, Heatmap profiling the expression of 23 genes involved in NF- κ B, inflammation and immune response. **f**, Western blot of p50, p65, phospho-p65 (p-p65) and the internal control (tubulin or histone 3) in THP-1^{PMA} cells. **g**, Relative mRNA and protein levels of IL-1 β and IL-6

in *TGR5* knockdown THP-1^{PMA} cells ($n = 6$). **h**, Western blot of pro-IL-1 β , *TGR5*, p50, p65, phospho-p65 and tubulin in *TGR5* knockdown THP-1^{PMA} cells. **i,j**, Groups of Abx-treated *TGR5* CKO mice pretreated with vehicle or TCDCa in drinking water for 4 weeks (**i**) or colonized with *A. muciniphila* for 48 h (**j**) were administered with anti-IFNAR1 IgG and subsequently intraperitoneally inoculated with SFTSV for the survival kinetics analysis. The two-sided P values were examined by Student's t test and data were presented as mean $\pm$ s.d. (**a,b,d,g**). The Kaplan–Meier method and log-rank test were used to analyse time-to-event data for treatment effect analysis (**i,j**).



Methods

Viruses, bacteria and cell culture

SFTSV strain HBMC16 was obtained from Wuhan Institute of Virology, Chinese Academy of Sciences (Wuhan, Hubei, China). The viral titre was determined by a focus-forming assay on Vero cells as described previously⁴³. *A. muciniphila* was obtained from American Type Culture Collection (ATCC, catalogue no. BAA835) and cultured in brain heart infusion medium (Oxoid) at 37 °C under anaerobic conditions. *L. reuteri* was purchased from the China Center for Type Culture Collection (CCTCC, catalogue no. AB 2014289) and cultured in De Man–Rogosa–Sharpe medium (Oxoid) at 37 °C under anaerobic conditions. *E. faecalis* was purchased from CCTCC (catalogue no. AB 2018154) and cultured in lysogeny broth medium (Oxoid) at 37 °C. The concentration of each bacterial species was quantified based on the optical density at 600 nm (OD₆₀₀).

THP-1 cells were purchased from ATCC and cultured in RPMI-1640 medium containing 10% fetal bovine serum (FBS; Gibco). Vero cells were obtained from ATCC and cultured in DMEM with 10% FBS. Huh-7 cells were obtained from Stem Cell Bank, Chinese Academy of Sciences and cultured with DMEM supplemented with 10% FBS. THP-1 cells were incubated with 100 ng ml⁻¹ phorbol-12-myristate 13-acetate (PMA; Sigma Aldrich) for 1 d to differentiate before utilization.

Mouse PBMCs were extracted by density-gradient centrifugation using mouse PBMC isolation kits (TBD) following the manufacturer's protocol. After isolation, PBMCs were cultured in RPMI-1640 medium supplemented with 10% heat-inactivated FBS at 37 °C.

MPHs were obtained as described previously⁴⁴. Briefly, the liver was perfused with calcium-free Hanks' buffered salt solution (Solarbio) supplemented with 0.5 mM EGTA, followed by DMEM supplemented with 100 collagen digestion unit. ml⁻¹ collagenase IV (Sigma). The digested liver was rapidly excised and gently shaken in F-12 medium containing 10% FBS and 1% penicillin/streptomycin to release the hepatocytes. Cells were then filtered through a 70-µm nylon filter into a 50-ml conical tube and washed twice with the same medium by centrifugation at 100g for 2 min. Finally, cells were seeded into a collagen-coated plate, left to stand for 4 h to allow attachment and washed once; serum-free medium was added for further experiments.

For gene depletion of *TGR5* or *FXR*, THP-1 cells at 40%–60% confluence were transfected with a set of siRNAs targeting *TGR5* (Santa Cruz Biotechnology, catalogue no. sc-61678) or *FXR* (Santa Cruz Biotechnology, catalogue no. sc-38848) using siRNA transfection medium (Santa Cruz Biotechnology, catalogue no. sc-36868) and siRNA transfection reagent (Santa Cruz Biotechnology, catalogue no. sc-29528) according to the manufacturer's instructions. Cells were used in experiments 6 h after transfection. Knockdown of *MyD88* or *TLR8* was performed by lentiviral transduction of THP-1 cells from Wuhan Institute of Virology.

Patients and sample collection

From May 2018 to August 2019, some 260 adult patients with laboratory-diagnosed SFTSV infection according to the guidelines released by the China Ministry of Health were recruited from the 990th Hospital in Xinyang city of Henan Province, the largest sentinel hospital for SFTS treatment in China. Patients with tumours, tuberculosis, diabetes or other infections (hepatitis virus, dengue virus, *Rickettsia* and *Borrelia*) were excluded. Clinical manifestations, laboratory test results and treatment regimens were retrieved from the medical records. In addition, 176 febrile patients without SFTSV infection and 19 healthy individuals who were living in the same areas as the patients during the study period were also recruited. Faecal and serum specimens were collected from patients at the acute and convalescent phases. Specimens were stored in study-provided sterile containers, kept at –20 °C and transferred to –80 °C upon return to the laboratory. The study was performed with the approval of the Ethical Committee of Beijing Institute of Microbiology and Epidemiology, and written informed consent was obtained from each participant.

Animals and viral infections

Six-week-old, sex-matched C57BL/6 mice purchased from the Model Animal Research Center of Nanjing University (Nanjing, China) were maintained in a specific-pathogen-free facility with a temperature- and humidity-controlled environment (22 ± 2 °C, 50% ± 10%), housed with a 12:12 h light/dark cycle and all animal experiments were strictly carried out in accordance with protocols approved (no. 117113) by the Institutional Animal Care and Use Committee of Zhejiang University. GF mouse experiments were carried out at the Laboratory Animal Center, Huazhong Agricultural University. Mice were freely fed an autoclavable diet (Wuxi Fanbo Biotechnology Co., catalogue no. M2011).

For all in vivo infection studies, mice were treated with anti-IFNAR1 IgG (1.5 mg) by intraperitoneal injection 1 d before intraperitoneal inoculation with 1,000 plaque forming unit (p.f.u.) of SFTSV in 100 µl of PBS. At the indicated time points, the lungs, livers and spleens were dissected, weighed, homogenized and titrated by qPCR. Standard cycling conditions, primers and probes were as described previously⁴⁵.

Abx treatment, faecal microbial transfer and bacterial colonization

Mice were administered an antibiotic cocktail containing ampicillin, neomycin, metronidazole and vancomycin via oral gavage for 5 d as described previously²⁰. Antibiotics were added to the drinking water, and animals were maintained with Abx- or PBS-containing water for the duration of the experiments.

For FMT experiments, faecal samples collected from SFTSV-infected mice at 3 d.p.i. were processed as described previously⁴⁶. In brief, faecal pellets were weighed and homogenized with sterile silica beads in 1 ml of PBS at 45 Hz for 1 min, filtered with a 100-µm strainer, and centrifuged at 6,000g for 15 min. The collected samples were resuspended in PBS with 10% (v/v) glycerol and frozen at –80 °C. At the end of the 14-d observation period of the survival study, the precollected faecal samples were divided into two groups based on whether the corresponding mice succumbed to or survived the infection. Faecal samples from each group were pooled, centrifuged and resuspended in PBS. Abx-treated mice were gavaged with 20 mg of pooled caecal contents in 100 µl of PBS 6 d after Abx oral administration.

For siRNA transfection experiments, mice were injected via the tail vein with 40 µg siRNA using in vivo jetPEI reagents (Polyplus).

For the *TGR5* conditional knockout (CKO) mouse protection experiment, groups of mice were given 1 mg of tamoxifen per 10 g (body weight) via intraperitoneal injections daily for 1 week, and an antibiotic treatment was performed in parallel via oral administration and drinking water, as described previously.

For the bacterial colonization and FMT experiments, Abx mice were subjected to gavage with 10¹⁰ colony-forming units of *A. muciniphila*, *L. reuteri*, *E. faecalis* or FMT in 200 µl of PBS 6 d post Abx oral administration. After 48 h of colonization, faecal samples were collected to determine the efficiency of colonization.

For pasteurization, *A. muciniphila* was inactivated by pasteurization for 30 min at 70 °C. The supernatants of *A. muciniphila* were centrifuged at 6,000g for 10 min at 4 °C and then passed through polyether-sulfone filters (0.22 µm; Merck Millipore) to remove the residual bacterial cells. WT B6 mice to oral gavage with *A. muciniphila* supernatant (200 µl per mouse) or pasteurized *A. muciniphila* (10⁹ colony-forming units per mouse) was initiated in parallel with antibiotic (Abx) treatment until the end of the experiments.

After *A. muciniphila*, *L. reuteri*, *E. faecalis*, FMT, *A. muciniphila* supernatant or pasteurized *A. muciniphila* treatments, mice were treated with anti-IFNAR1 IgG (1.5 mg), inoculated with 1,000 p.f.u. of SFTSV and observed for mortality for 14 d.

Animal protection study with metabolites

For the TCDCA (MedChemExpress) protection experiment, 3 mM TCDCA was added to the drinking water for 4 weeks before infection

with SFTSV, and antibiotic treatment was performed 1 week before SFTSV infection. For the acetate or propionate (Sigma) protection experiment, 200 mM acetate or propionate was added to the drinking water for 1 week before infection with SFTSV, and antibiotic treatment was performed 1 week before SFTSV infection. For the HAL (Selleck) protection study, groups of mice were given 10 mg of HAL per kg (body weight) by oral gavage daily for 1 week, and antibiotic treatment was performed in parallel via oral administration and drinking water as described previously. For the TGR5 CKO mouse protection experiment, 3 mM TCDCA was added to the drinking water for 4 weeks before infection with SFTSV, and antibiotics and tamoxifen (dissolved in corn oil, 1 mg per 10 g (body weight) daily by intraperitoneal injection) treatment was performed 1 week before SFTSV infection. After TCDCA, HAL, acetate or propionate treatments, mice were treated with anti-IFNAR1 IgG (1.5 mg), inoculated with 1,000 p.f.u. of SFTSV and observed for mortality for 14 d.

Faecal bacteria quantification

Faecal bacterial quantification was determined by qPCR (primers used are listed in the table of Key Resources). The faecal bacterial DNA was isolated using a TIANamp Stool DNA Kit (TIANGEN). qPCR was performed using SYBR Green Real-time PCR Master Mix (TOYOBO).

Cytokine expression analysis

Total RNA from bead-homogenized tissue samples or cell culture was extracted using TRIzol reagent (Invitrogen) following the manufacturer's instructions. The cytokine level was determined by quantitative PCR with reverse transcription using the HiScript II One Step qRT-PCR SYBR Green Kit (Vazyme).

IL-1 β and IL-6 protein levels in serum samples and culture supernatants were measured by the corresponding enzyme-linked immunosorbent assay (ELISA) kits (MultiSciences) following the manufacturer's instructions.

Western blot analysis

Tissues and cells treated as indicated were lysed with RIPA lysis buffer (Beyotime). The lysates were subjected to 10% SDS-polyacrylamide gel electrophoresis and then transferred to polyvinylidene difluoride membranes (Millipore). Proteins were further incubated with the indicated primary antibodies and then horseradish peroxidase-conjugated secondary antibodies. Protein bands were probed using an enhanced chemiluminescence kit (Vazyme) with a ChemiDoc Touch Gel Imaging System (Bio-Rad).

Tissue histology and staining

Livers, lungs and spleens soaked in 4% paraformaldehyde solution were dehydrated, embedded in paraffin, cut into 4- μ m thick sections, and stained with haematoxylin and eosin (HE) using standard procedures.

Deparaffinized spleen sections were blocked with 10% normal goat serum for 30 min and incubated with anti-IL-6 (Proteintech) polyclonal antibody overnight at 4 °C. Sections were incubated with FITC-conjugated anti-rabbit secondary antibody for 55 min at room temperature, microwaved and subjected to Nucleocapsid protein N (NP) staining with anti-NP polyclonal antibody (4 °C overnight) and cy3-conjugated anti-rabbit secondary antibody (room temperature, 55 min). Finally, the sections were incubated with 4',6-diamidino-2-phenylindole solution at room temperature for 10 min. Microscopy was performed, and images were collected by fluorescence microscopy.

DNA extraction, 16S rRNA amplicon sequencing and data analyses

Faecal samples (~200 mg) were resuspended in Qiagen's ASL buffer and homogenized for 2 min. Total faecal DNA was extracted from the supernatant using a QIAamp DNA Stool Mini Kit (Qiagen). DNA

concentration and purity were measured by Qubit (Thermo Fisher Scientific). The stool DNA was then amplified using Phusion High-Fidelity PCR Master Mix (New England Biolabs) by PCR targeting the variable regions 3 and 4 (V3–V4) of the 16S rRNA gene (forward primer, ACTC-CTACGGGAGGCAGCA; reverse primer, GGACTACHVGGGTWTCTAAT). Multiplex sequencing of amplicons with sample-specific barcodes was performed using an Illumina NovaSeq platform. Paired-end reads were merged into long sequences using FLASH v.1.2.7, a very fast and accurate analysis tool designed to merge paired-end reads when there are overlaps between reads1 and reads2 (ref. ⁴⁷). The merged sequences were then analysed using the QIIME v.1.9.1 software package⁴⁸.

High-throughput amplicon sequencing of the full-length 16S rRNA gene

Full-length 16S rRNA gene amplification (forward primer, AGAGTTT-GATCCTGGCTCAG; reverse primer, GNTACCTTGTTACGACTT) was conducted on the DNA samples with TransStart FastPfu DNA Polymerase (TransGen Biotech). Gel electrophoresis was performed for gel-based size selection. The gels were then extracted with a QIAquick Gel Extraction Kit (Qiagen). Barcoded amplicons were pooled for multiplex sequencing and library construction with the SMRTbell Template Prep Kit (PacBio) following the procedure. Sequencing was performed by the PacBio Sequel platform.

Bacterial supernatant filtrate preparation

The supernatant was collected from a 100 ml culture of *A. muciniphila* and filtered using filters with a number of differing pore sizes and molecular mass cut-offs. Briefly, the supernatants were centrifuged at 6,000g and 4 °C for 10 min and then passed through polyether-sulfone filters (0.22 μ m; Merck Millipore) to remove the residual bacterial cells. The supernatants were passed through 100, 50 and 10 kDa filters (Merck Millipore) at 3,200g for 10 min. Each filtrate was frozen at –80 °C until assayed. The protein content of the filtrates was determined by using BCA assay kits (Beyotime).

LC–MS/MS (quasi-targeted metabolomics)

Metabolites were extracted from the 10 kDa bacterial supernatant samples or serum. Briefly, the samples were resuspended in pre-chilled 80% methanol by vortexing. The samples were then incubated on ice for 5 min and centrifuged at 15,000g and 4 °C for 15 min. The supernatant was injected into the LC–MS/MS system for analysis. Ultrahigh-performance liquid chromatography coupled to tandem mass spectrometry (UHPLC–MS/MS) analyses were performed using a Vanquish UHPLC system (Thermo Fisher) coupled with an Orbitrap Q Exactive HF-X mass spectrometer (Thermo Fisher) by Novogene. The raw data files generated by UHPLC–MS/MS were processed using Compound Discoverer 3.1 (CD3.1, Thermo Fisher) to perform peak alignment, peak picking and quantitation for each metabolite. These metabolites were annotated using the KEGG database (<https://www.genome.jp/kegg/pathway.html>), HMDB database (<https://hmdb.ca/metabolites>) and LIPIDMaps database (<http://www.lipidmaps.org/>). Principal component analysis and partial least-squares discriminant analysis were performed at metaX (a flexible and comprehensive software for processing metabolomics data).

LC–MS/MS determination of serum BAs and HAL concentrations

Serum samples (100 μ l) were resuspended in 500 μ l of acetonitrile/methanol (8:2) and centrifuged at 12,000g for 20 min. The supernatant was then dried using a nitrogen blower. The precipitates were reconstituted with 100 μ l of water/acetonitrile (8:2) with formic acid (0.1%) by thorough vortexing and centrifugation. Finally, the supernatant (2 μ l) was injected into the LC–MS/MS system for analysis. A UHPLC–MS/MS system (ExionLC AD UHPLC-QTRAP 6500+, AB SCIEX Corp.) was used to quantitate bile acids and HAL at Novogene. Liquid chromatography–mass spectrometry was used to detect the

concentration series of the standard solution. The concentration of the standard was used as the abscissa, and the ratio of the internal standard peak area was used as the ordinate to investigate the linearity of the standard solution.

RNA interference

siRNAs were transfected into MPHs or human THP-1 cell lines with Lipofectamine 3000 reagent (Invitrogen) or siRNA Transfection Reagent (Santa Cruz Biotechnology, catalogue no. sc-29528) following the manufacturer's instructions. Human TGR5- and FXR-specific siRNAs were designed and synthesized by Santa Cruz Biotechnology, and mouse Baat-specific siRNAs were designed and synthesized by GenePharma. The efficiency of interference was determined by qPCR or western blotting.

Transcriptomics analysis

THP-1^{PMA} cells, which were pretreated with 100 μ M MGCDCA then infected with SFTSV (MOI1) for 24 h, were collected and total RNA was extracted with RNAiso Plus (TAKALA) and assessed with the Agilent 4200 system (Agilent Technologies), Qubit 3.0 (Thermo Fisher Scientific) and Nanodrop One (Thermo Fisher Scientific) at the same time. RNA-seq libraries were generated and sequenced by Guangdong Magigene Biotechnology. Triplicate samples of all assays were constructed in an independent library, and the following sequencing and analysis was performed. Whole messenger RNA-seq libraries were generated using NEB Next Ultra Nondirectional RNA Library Prep Kit for Illumina (New England Biolabs) following the manufacturer's recommendations. Clustering of the index-coded samples was performed on a cBot Cluster Generation System. After cluster generation, the library was sequenced on an Illumina NovaSeq 6000 platform and 150 bp paired-end reads were generated. Raw data of the fastq format were processed by Trimmomatic (v.0.36) to acquire clean data (clean reads). Clean reads were mapped to NCBI Rfam databases to remove the rRNA sequences by Bowtie2 (v.2.33). The remaining mRNA sequences were mapped to reference genome by Hisat2 (2.1.0). HTSeq-count (v.0.9.1) was used to obtain the read count and function information of each gene according to the result of the mapping. Differentially expressed genes of two conditions/groups was performed using edgeR (v.3.16.5). GO analysis of differentially expressed genes was implemented using clusterProfiler (v.3.4.4), in which gene-length bias was corrected.

Statistics and reproducibility

Statistical analyses were performed with Prism GraphPad software v.8.0. Error bars represent standard errors of the means in all figures, and *P* values were determined by two-tailed Student's *t* test or analysis of variance. *R*² was estimated for the correlation analysis of two continuous variables. The Kaplan–Meier method and log-rank test were used to analyse time-to-event data for treatment effect analysis. A two-sided *P* value <0.05 was considered statistically significant. No data points were excluded from the analysis. Data collection and analysis were not blinded. Data distribution was assumed to be normal but this was not formally tested.

All the experiments were replicated and the number of replicates is stated in the figure legends. Representative images for HE staining, indirect fluorescent assay (IFA) and western blotting are from at least *n* = 3 independent sample preparations.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

16S rRNA sequence data are available in the Sequence Read Archive (SRA) under BioProject accession no. [PRJNA890424](https://www.ncbi.nlm.nih.gov/bioproject/PRJNA890424) and [PRJNA888451](https://www.ncbi.nlm.nih.gov/bioproject/PRJNA888451). RNA-seq data are available in the SRA under BioProject accession no.

[PRJNA889171](https://www.ncbi.nlm.nih.gov/bioproject/PRJNA889171). Source data can be accessed on FigShare: <https://doi.org/10.6084/m9.figshare.21334422>. Source data are provided with this paper.

Code availability

No custom code was used.

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Author contributions

S.J.Z. and W.L. designed the experiments. J.X., H.L., X.Z., T.Y., M.Y., Y.Z. and S.C. performed the experiments. J.X., H.L., X.Z. and T.Y. conducted the bioinformatics analysis. N.C. and C.Y. collected samples and data. J.L. commented on and revised the drafts of the manuscript. S.J.Z., W.L., J.X., H.L. and T.Y. wrote the paper. S.J.Z. and W.L. supervised the research, coordination and strategy.

Competing interests

The authors declare no competing interests.

Additional information

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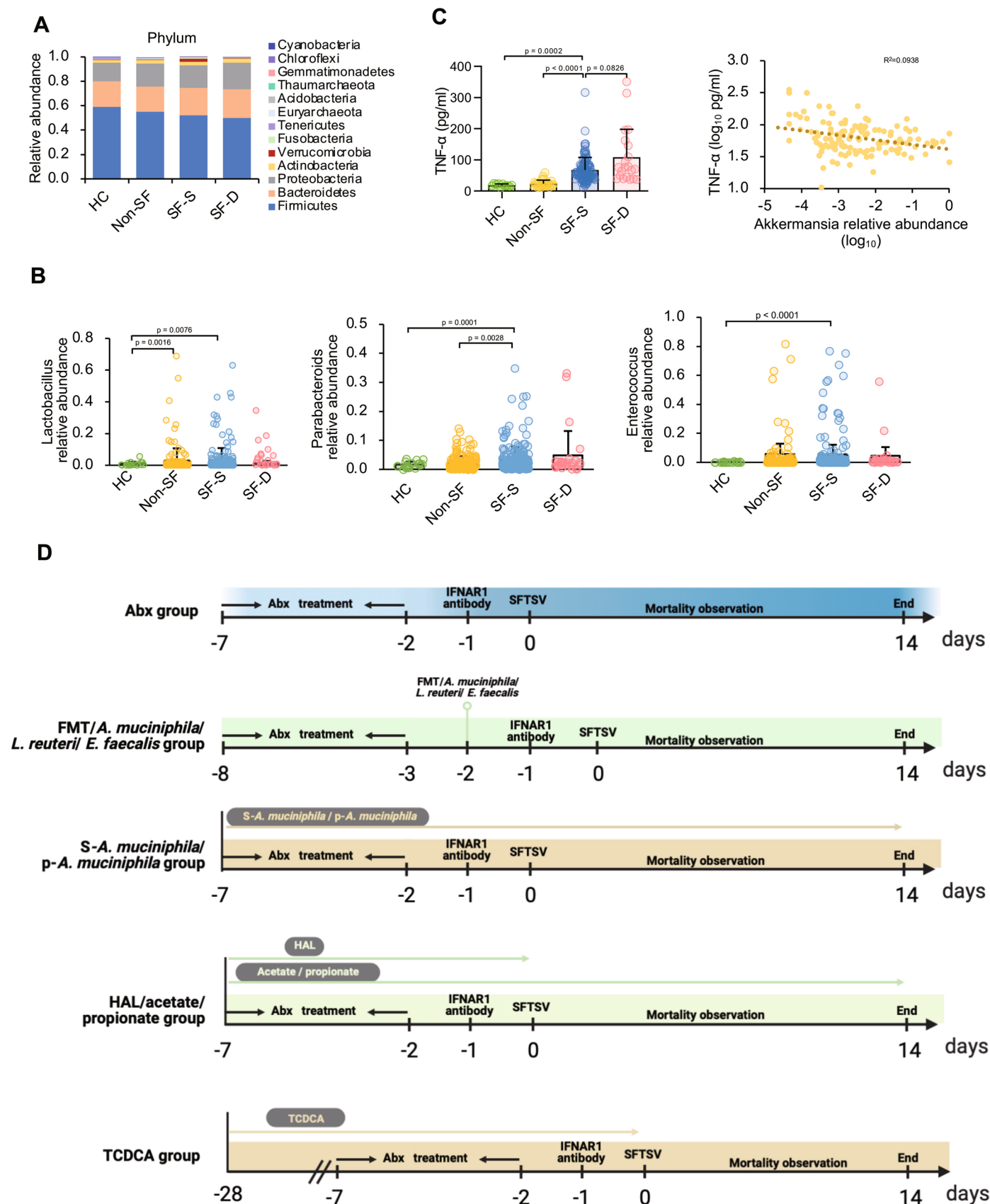
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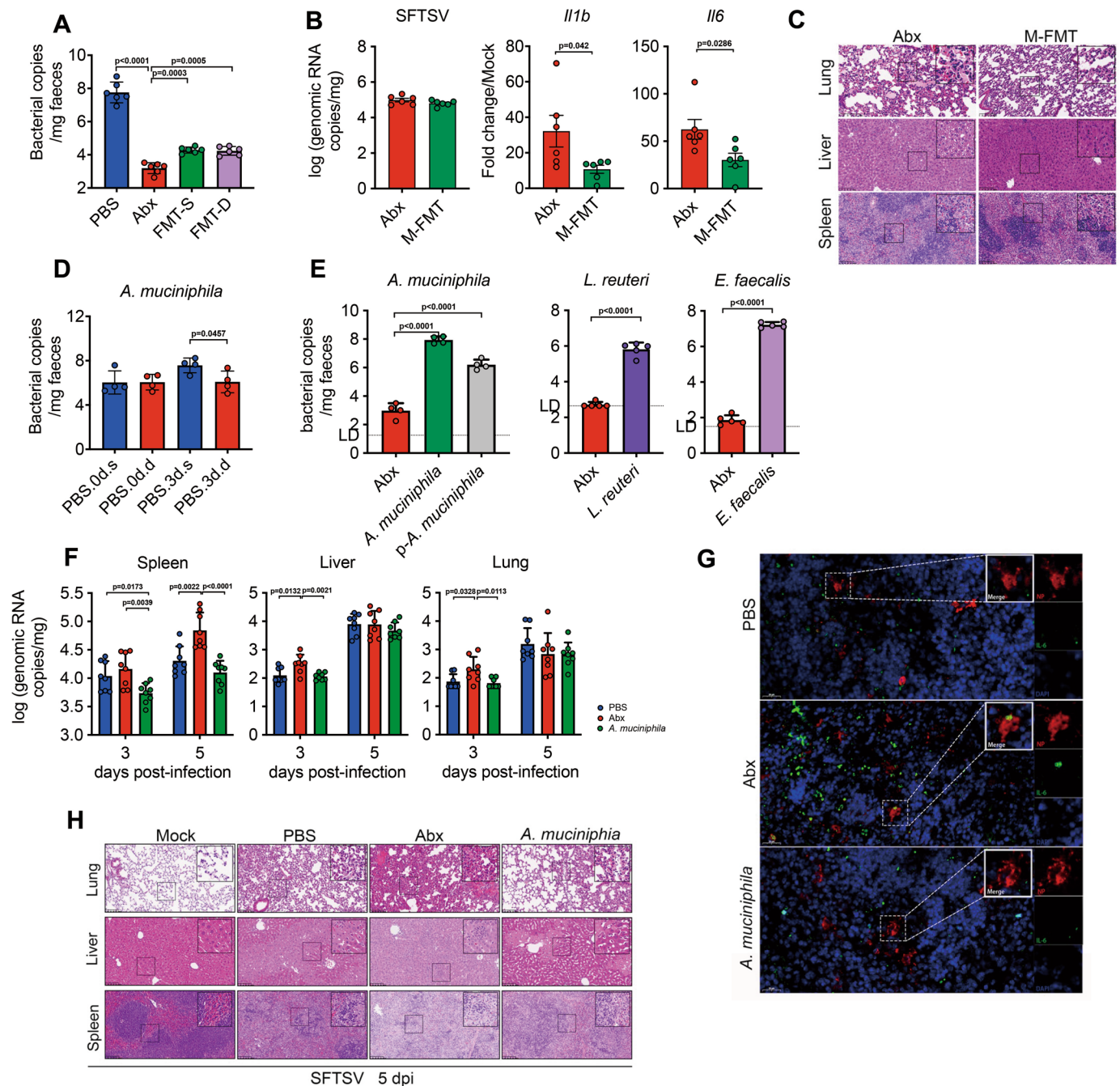
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Extended Data Fig. 1 | See next page for caption.

Extended Data Fig. 1 | The intestinal microbiota of surviving SFTSV-infected individuals. (a) Relative abundance of phyla among the four groups. **(b)** Relative abundance of *Lactobacillus*, *Parabacteroides* and *Enterococcus* among the four groups (HC n = 19, Non-SF n = 176, SF-S n = 233, SF-D n = 27). **(c)** Serum concentration of TNF- α among the four groups and the association with the relative abundance of *Akkermansia* (HC n = 10, non-SF n = 25, SF-S n = 107, SF-D

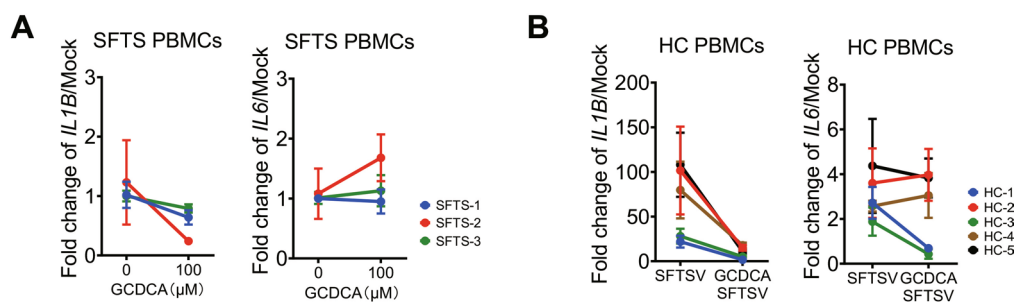
n = 21). HC, healthy controls; Non-SF, febrile patients without SFTSV infection; SF-S, surviving patients with SFTSV infection; SF-D, deceased patients with SFTSV infection. **(d)** Schematic representation of treatment with Abx, FMT, bacteria, HAL, acetate, propionate or TCDCA. The two-sided *P* values were examined by Student's *t* test and data were presented as mean values $\pm$ SD (**b**, and **c**). R^2 and exact two-sided *P* values calculated by Pearson's test are shown (**c**).



Extended Data Fig. 2 | Microbiota reconstitution with *A. muciniphila* prevents SFTSV early infection and inhibits resulting systemic inflammatory responses.

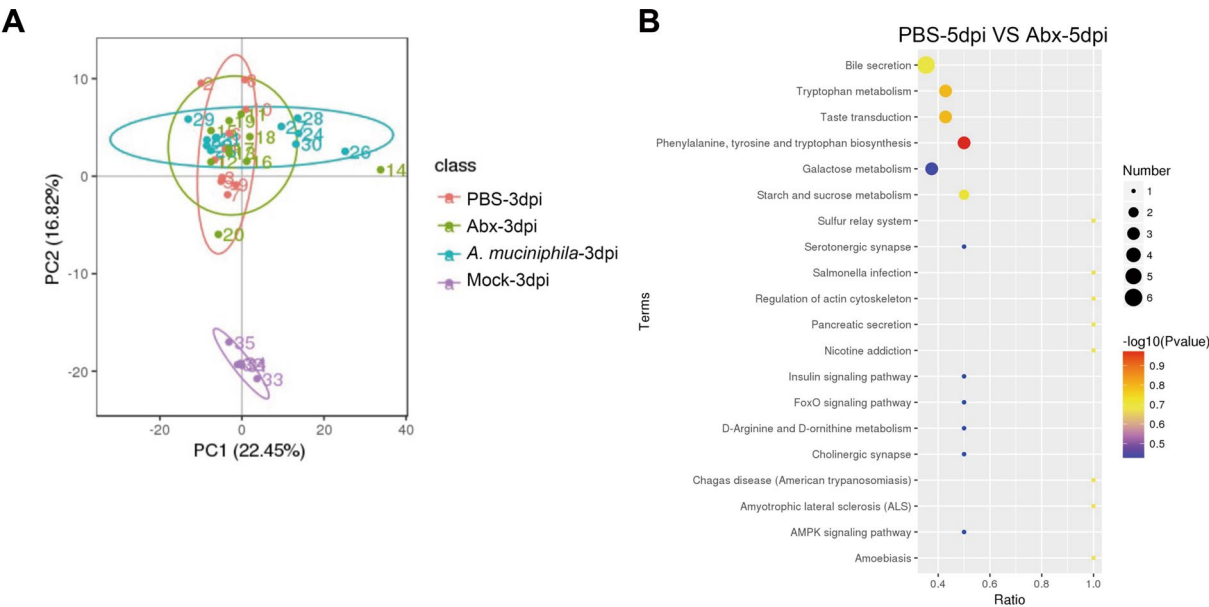
(a) qPCR of 16 s rRNA genomic copies in faeces from PBS-treated, Abx-treated or FMT-transferred mice at 2 days after transfer ($n = 6$). (b) qPCR of *IL-1 β* and *IL-6* mRNA (right) and SFTSV RNA (left) in spleens from Abx-treated mice with or without FMT (faeces from recovered SFTSV mice) and infected with SFTSV at 3 days post-infection ($n = 6$). (c) H. E. staining of lung, liver or spleen cross sections from Abx-treated mice with or without FMT (faeces from recovered SFTSV mice) and infected with SFTSV at 3 dpi. Boxed areas are magnified immediately in the top right corner. (d) qPCR of *A. muciniphila* genomic copies in faeces from SFTSV infected mice at 0- and 3-days post infection ($n = 4$). (e) qPCR of *A. muciniphila*, *L. reuteri* or *E. faecalis* genomic

copies in faeces from Abx-treated or bacteria-colonized mice at 2 days post colonization (*A. muciniphila* $n = 4$, *L. reuteri* and *E. faecalis* $n = 5$). (f) qPCR of SFTSV RNA in the spleen, liver or lung from PBS-treated, Abx-treated or *A. muciniphila*-colonized mice infected with SFTSV at 3 and 5 dpi ($n = 8$). (g) IFA of spleen sections from Abx-treated mice colonized with or without *A. muciniphila* and infected with SFTSV at 3 dpi. SFTSV protein NP and IL-6 protein were double stained with the respective antibodies. (h) H. E. staining of lung, liver or spleen cross sections from Abx-treated mice colonized with or without *A. muciniphila* and infected with SFTSV at 5 dpi. Boxed areas are magnified immediately in the top right corner. The two-sided P values were examined by Student's t test and data were presented as mean values $\pm$ SD (a, b, d, e and f).



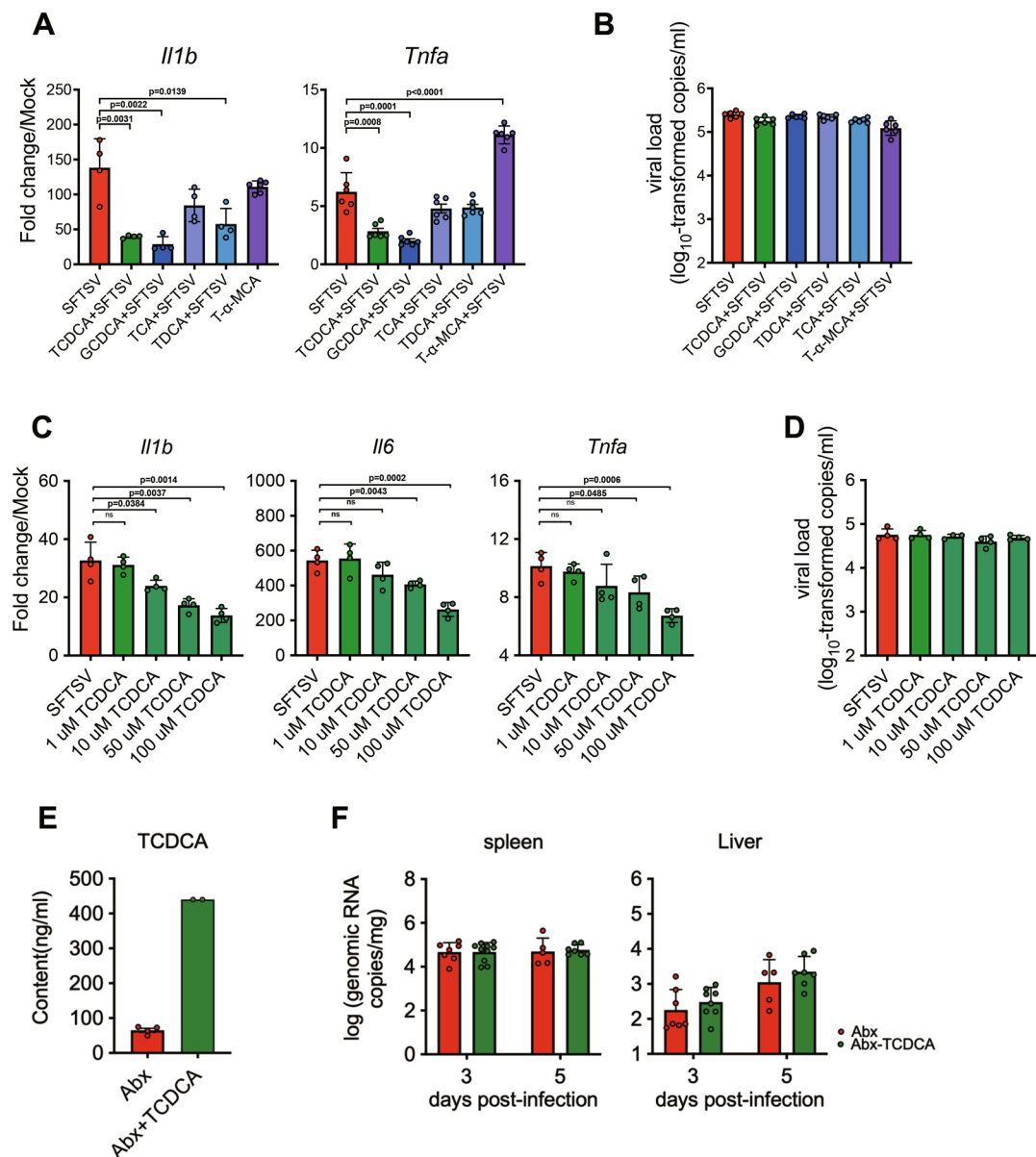
Extended Data Fig. 3 | *A. muciniphila*-associated conjugated primary bile acids GCDCA and TCDCA suppress SFTSV-induced inflammatory responses in human PBMCs in vitro. (a) Relative mRNA levels of IL-1 β and IL-6 in SFTS

patients (n = 3) and **(b)** healthy donor PBMCs treated with GCDCA and then infected with SFTSV (MOI1) at 24 hpi (n = 6). Data were presented as mean values $\pm$ SD **(a and b)**.



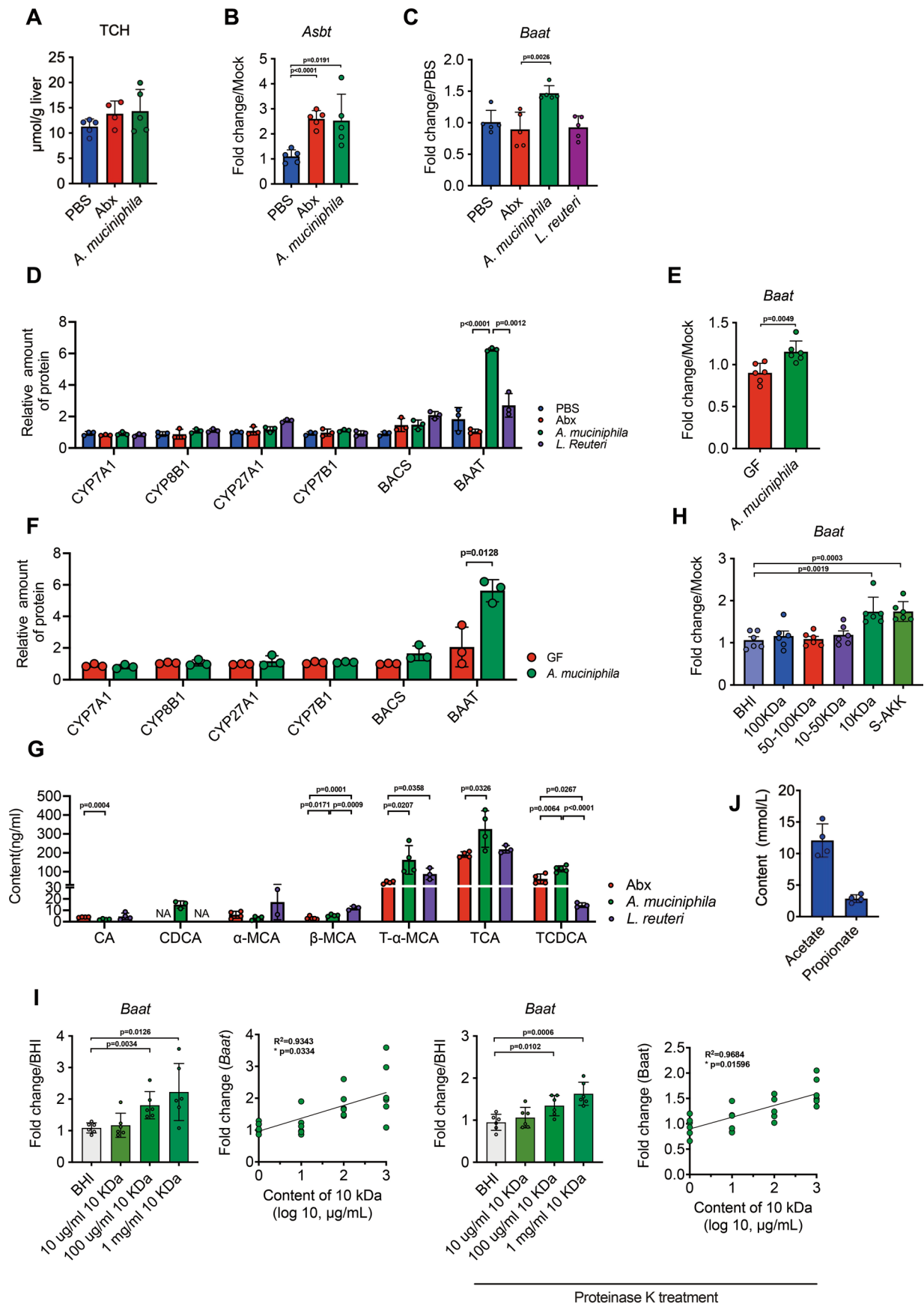
Extended Data Fig. 4 | *A. muciniphila* colonization in antibiotic-pretreated mice upregulates TCDCA secretion. (a) PCA of serum from four groups: PBS-treated, Abx-treated and *A. muciniphila*-colonized mice infected with SFTSV

at 3 dpi and nontreated mice. **(b)** KEGG analysis of differentially regulated metabolites between the PBS group and the Abx group. The two-sided *P* values were examined by Student's *t* test.



Extended Data Fig. 5 | TCDCA protects Abx mice by prohibiting SFTSV-induced systemic inflammation. (a) qPCR of IL-1 β or TNF- α mRNA and **(b)** SFTSV RNA in mouse PMBCs pretreated with TCDCA, GCDCA, TCA, TDCA or T- α -MCA and then infected with SFTSV (MOI1) at 24 hpi (IL-1 β n = 4, TNF- α and SFTSV RNA n = 6). **(c)** Relative mRNA transcripts of proinflammatory cytokines in SFTSV-infected mouse PBMCs treated with different doses of TCDCA and **(d)**

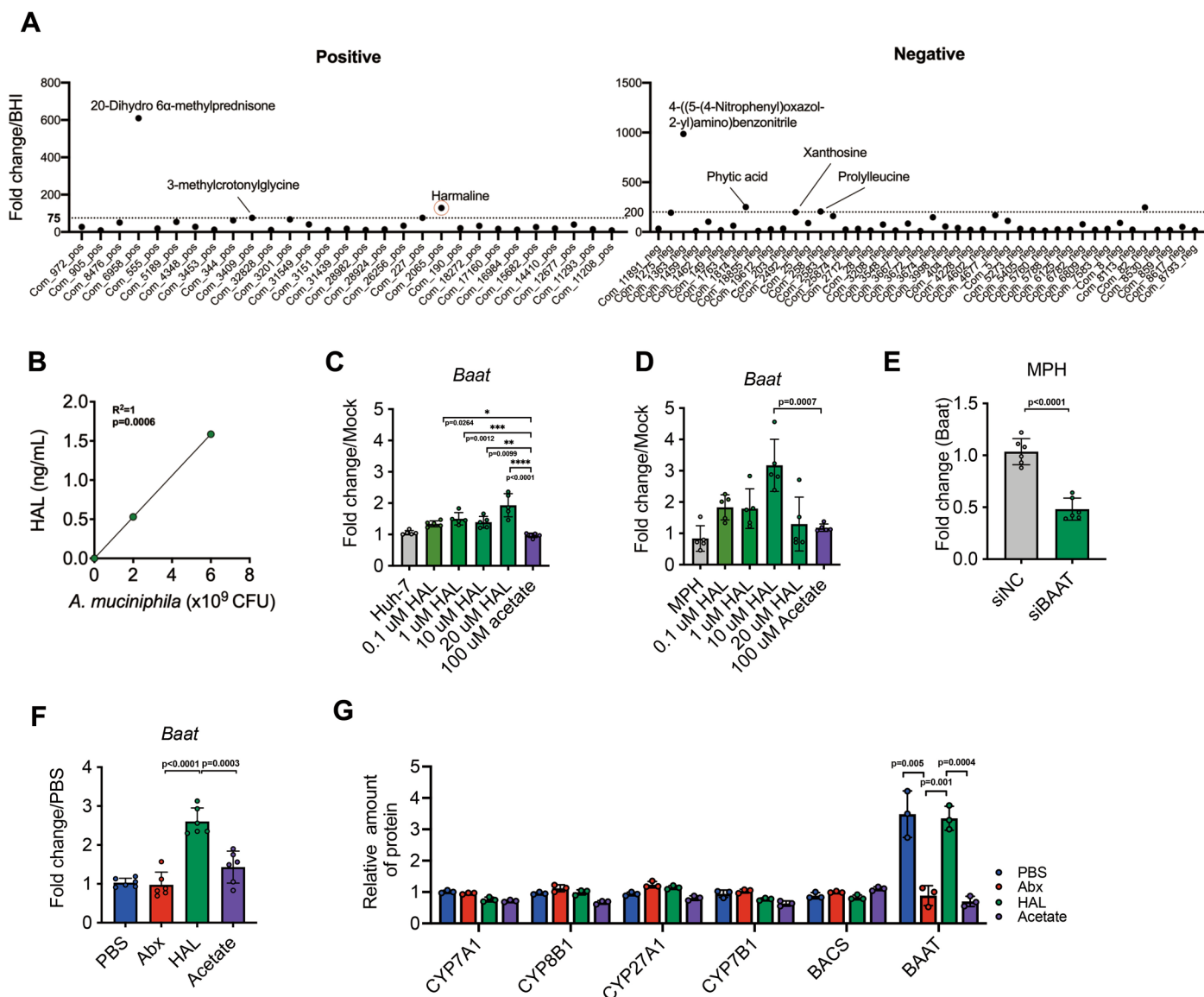
respective viral loads at 24 hpi (n = 4). **(e)** Serum concentration of TCDCA in Abx mice treated with TCDCA (Abx n = 4, TCDCA n = 2). **(f)** qPCR of SFTSV RNA in the spleen and liver from PBS-treated and Abx-treated mice treated with or without TCDCA and infected with SFTSV at 3 and 5 dpi (Abx 3dpi n = 7, Abx 5dpi n = 5, TCDCA 3dpi n = 8, TCDCA 5dpi n = 7). The two-sided *P* values were examined by Student's *t* test (**a**, and **c**). Data were presented as mean values $\pm$ SD (**a** to **f**).



Extended Data Fig. 6 | See next page for caption.

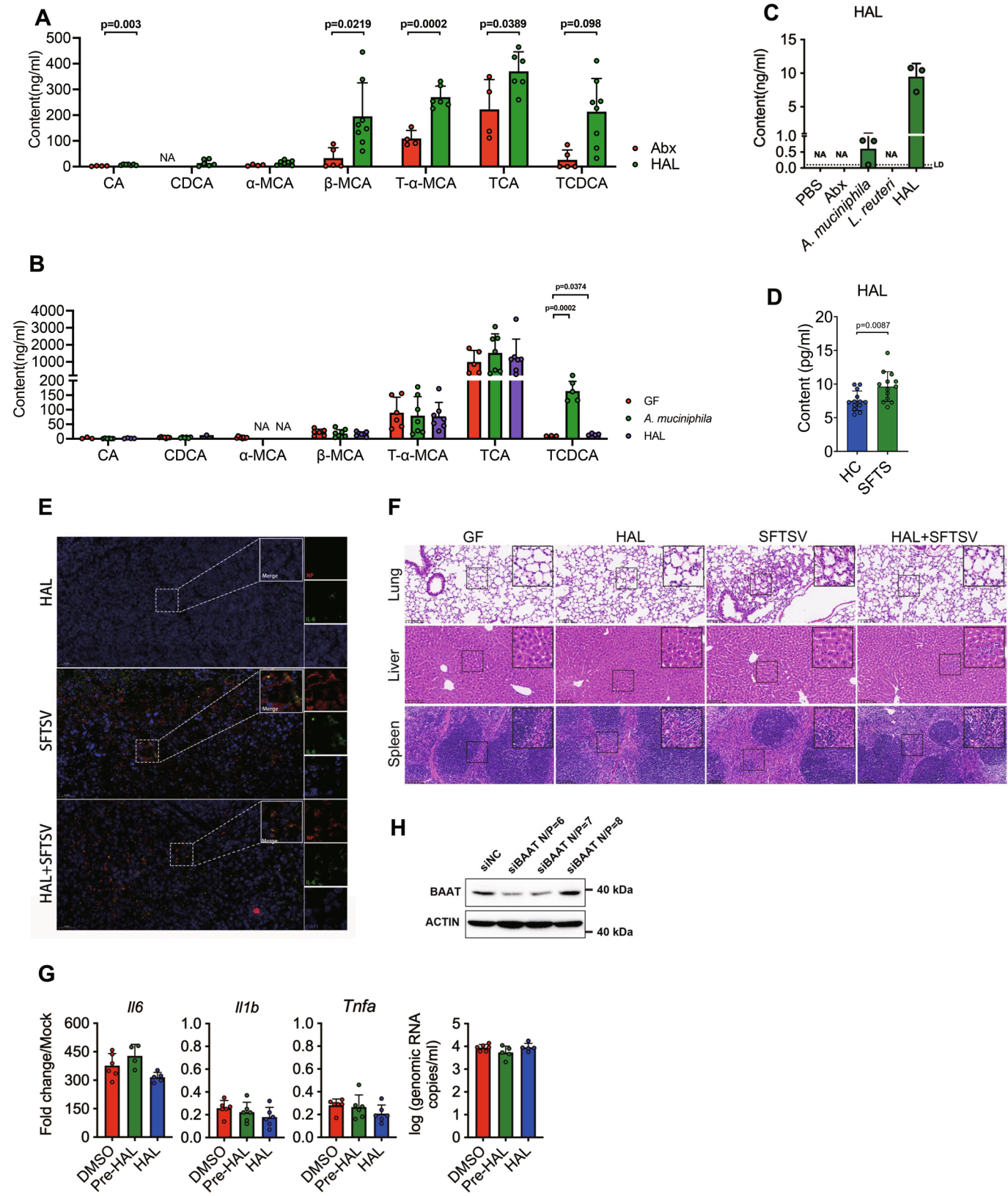
Extended Data Fig. 6 | *A. muciniphila*-associated metabolites enhance the expression of BAAT both *in vivo* and *in vitro*. (a) Total hepatic cholesterol level of PBS-treated, Abx-treated or *A. muciniphila*-colonized mice at 5 days post-colonization (n = 5). (b) qPCR of *Asbt* mRNA in the ileum of PBS-treated, Abx-treated or *A. muciniphila*-colonized mice at 5 days post-colonization (n = 5). (c) qPCR of BAAT mRNA in liver from PBS-treated, Abx-treated, *A. muciniphila*-colonized or *L. reuteri*-colonized mice at 5 days post-colonization (n = 5). (d) Quantification of the proteins in Fig. 5a. The protein levels of enzymes in the bile acid biosynthesis pathways were normalized to those of ACTIN. The amount of each protein in the fasted state was defined as 1 (n = 3). (e) qPCR of BAAT mRNA in liver from GF mice colonized with or without *A. muciniphila* at 5 days post-colonization (n = 6). (f) Quantification of the proteins in Fig. 5b. The protein

levels of enzymes in the bile acid biosynthesis pathways were normalized to those of TUBULIN. The amount of each protein in the fasted state was defined as 1 (n = 3). (g) Serum bile acid concentration of Abx-treated, *A. muciniphila*-colonized or *L. reuteri*-colonized mice at 5 days post colonization (n = 4). (h) qPCR of BAAT mRNA in Huh-7 cells treated with different fractions of *A. muciniphila* culture supernatants at 24 h (n = 6). (i) qPCR of BAAT mRNA in Huh-7 cells treated with different doses of 10 kDa or less filtrate with or without proteinase K digestion and the correlations with the content of 10 kDa or less filtrate (n = 6). (j) The content of acetate or propionate in the supernatant of *A. muciniphila* (n = 4). The two-sided *P* values were examined by Student's *t* test and data were presented as mean values $\pm$ SD (a–j).



Extended Data Fig. 7 | The *A. muciniphila* metabolite harmaline upregulates the expression of BAAT. (a) Relative abundance changes of positive or negative metabolites in 10 kDa or less filtrate of *A. muciniphila* cultured supernatants. (b) The correlations between the content of HAL and *A. muciniphila* cell numbers. (c, d) qPCR of BAAT mRNA in (c) Huh-7 or (d) MPH cells treated with different doses of HAL or butyrate at 24 h ($n=5$). (e) qPCR of BAAT mRNA in MPH cells transfected with siNC or siBAAT at 24 h ($n=6$). (f) qPCR of BAAT mRNA in liver

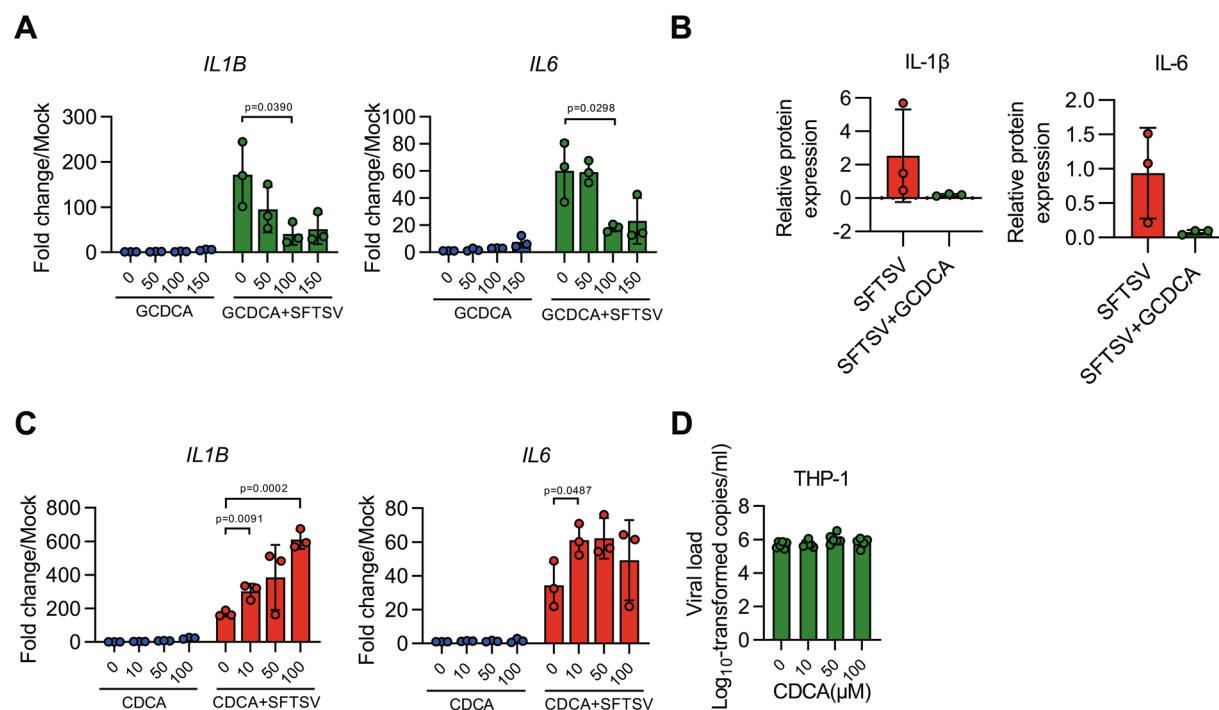
from PBS-treated, Abx-treated, HAL-treated or acetate-treated mice at 5 days post-treatment ($n=6$). (g) Quantification of the proteins in Fig. 5g. The proteins of the bile acid biosynthesis pathway were normalized to that of ACTIN. The amount of each protein in the fasted state was defined as 1 ($n=3$). The two-sided P values were examined by Student's t test and data were presented as mean values $\pm$ SD (c–g).



Extended Data Fig. 8 | See next page for caption.

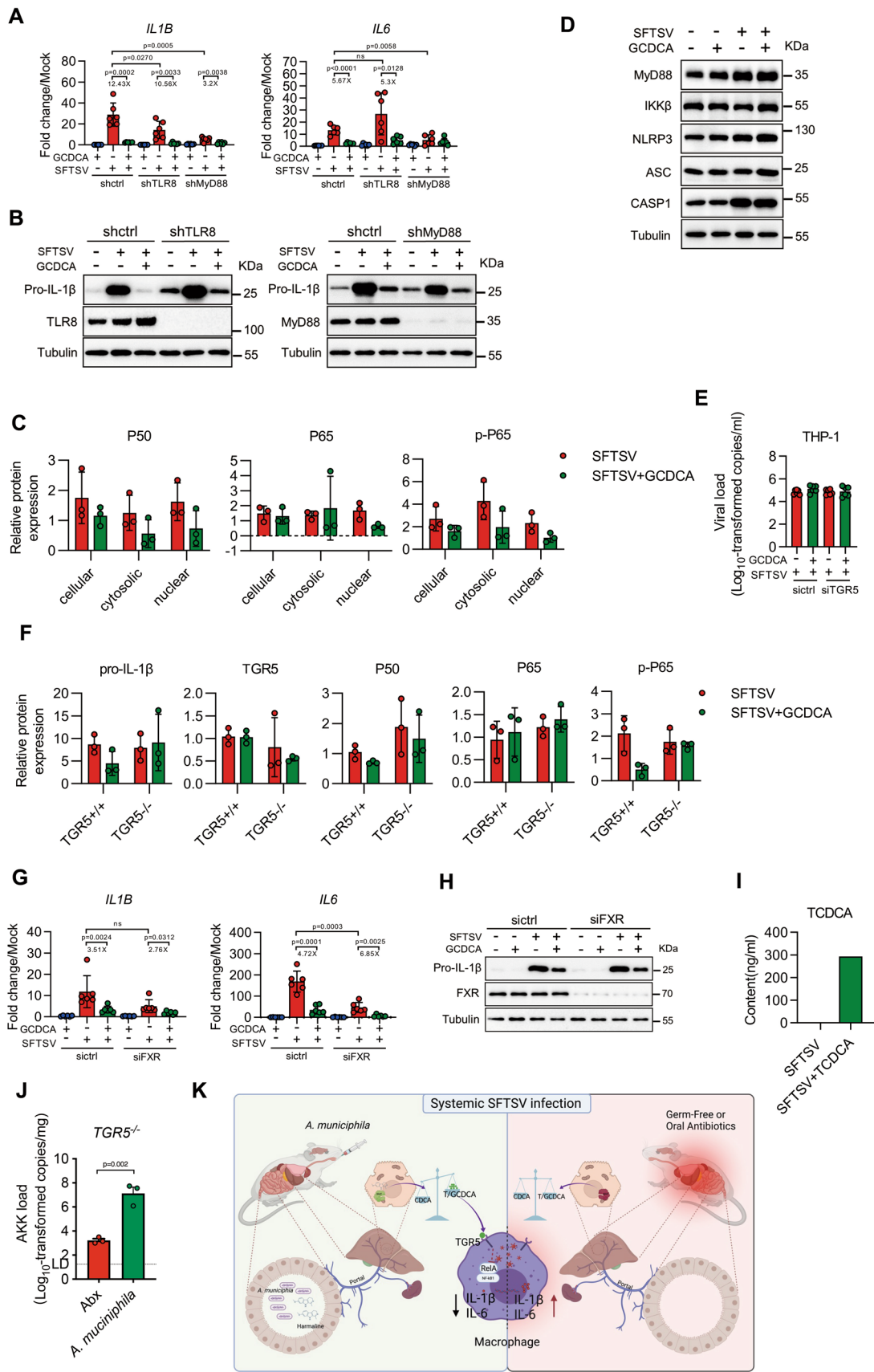
Extended Data Fig. 8 | The *A. muciniphila* metabolite harmaline prohibits systemic inflammatory responses in Abx-treated and GF mice. (a) Serum bile acid concentration of Abx- and HAL-treated mice and (b) GF mice treated with or without *A. muciniphila* or HAL at 5 days post-colonization (n = 8). (c) Serum HAL concentration of PBS-treated, Abx-treated, *A. muciniphila*-colonized, *L. reuteri*-colonized, HAL-treated mice at 2 days post colonization (n = 3). (d) Serum HAL concentrations of recovered patients with SFTSV infection and recovered febrile patients without SFTSV infection. Serum samples were collected approximately two weeks after symptom onset (n = 13). (e) IFA images of spleen sections from GF mice treated with or without HAL at 3 dpi. SFTSV protein NP and IL-6 protein were

double stained with the respective antibodies. (f) H. E staining of lung, liver or spleen cross sections from GF mice treated with or without HAL and infected with SFTSV at 3 dpi. Boxed areas are magnified immediately in the top right corner. (g) Relative mRNA levels of IL-1 β , IL-6, TNF- α (left) and SFTSV RNA (right) in mouse PBMCs that were pretreated with 10 μ M HAL and then infected with SFTSV (MOI 1) at 24 hpi or HAL was added to the PBMCs simultaneously with SFTSV infection (n = 6). (h) Expression of Baat and actin in the livers of Abx-treated wild-type B6 mice transiently transfected with *siNC* or *siBAAT* (N/P ratio is 6 to 8). The two-sided *P* values were examined by Student's *t* test (a, b and d). Data were presented as mean values $\pm$ SD (a–d and g).



Extended Data Fig. 9 | GCDCA suppresses SFTSV-induced inflammatory cytokine expression in THP-1 cells in a dose-dependent manner. (a) Relative mRNA levels of IL-1 β and IL-6 in THP-1^{PMA} cells that were pretreated with 50, 100, or 150 μ M GCDCA and then infected with SFTSV (MOI1) at 24 hpi ($n = 3$). (b) Relative protein levels of IL-1 β and IL-6 in THP-1^{PMA} cells pretreated with 100 μ M

GCDCA and then infected with SFTSV (MOI1) at 24 hpi ($n = 3$). (c) Relative mRNA levels of IL-1 β , IL-6 ($n = 3$) and (d) SFTSV RNA in THP-1^{PMA} cells ($n = 6$) pretreated with 10, 50, or 100 μ M CDCA and then infected with SFTSV (MOI1) at 24 hpi. The two-sided P values were examined by Student's t test (a and c). Data were presented as mean values $\pm$ SD (a to d).



Extended Data Fig. 10 | See next page for caption.

Extended Data Fig. 10 | GCDCA does not hinder SFTSV-induced inflammatory responses via the TLR8-MyD88 or NLRP3-inflammasome signalling pathways. THP-1^{PMA} cells were pretreated with 100 μ M GCDCA and then infected with SFTSV (MOI1) for 24 hours. **(a)** Relative mRNA levels of *IL1B* and *IL6* in TLR8- or MyD88-knockdown THP-1^{PMA} cells (n = 6). **(b)** Western blot of pro-IL-1 β , TLR8, MyD88 and tubulin in TLR8- or MyD88-knockdown THP-1^{PMA} cells. **(c)** Relative protein levels of P50, P65 and phos-P65 in THP-1^{PMA} cells (n = 3). **(d)** Western blot of MyD88, IKK β , NLRP3, ASC, CASP1 and tubulin in THP-1^{PMA} cells. **(e)** SFTSV RNA level in *TGR5* knockdown THP-1^{PMA} cells (n = 6). **(f)** Relative protein levels of pro-IL-1 β , TGR5, p50, p65 and phos-p65 in TGR5 knockdown THP-1^{PMA} cells (n = 3). **(g and h)** THP-1^{PMA} cells were pretreated with 100 μ M GCDCA and then

infected with SFTSV (MOI1) for 24 hours. **(g)** Relative mRNA levels of *IL1B* and *IL6* in FXR knockdown THP-1^{PMA} cells (n = 6). **(h)** Western blot of pro-IL-1 β , FXR and tubulin in FXR knockdown THP-1^{PMA} cells. **(i)** Serum TCDCA concentration of Abx-treated TGR5 CKO mice with or without TCDCA treatment at 5 dpi. **(j)** qPCR of *A. muciniphila* genomic copies in faeces from Abx-treated or bacteria-colonized mice at 2 days post colonization (n = 3). **(k)** Schematic illustration of the *Akkermansia muciniphila* metabolite harmaline protecting against severe fever with thrombocytopenia syndrome through the bile acid-TGR5-NF- κ B axis. The two-sided *P* values were examined by using Student's *t* test (**a, g and j**). Data were presented as mean values $\pm$ SD (**a, c, e–g and j**).

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

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| ☐ | ☒ | The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement |
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| ☐ | ☒ | A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ | For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ | For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ | For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☐ | ☒ | Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection No software was used

Data analysis FLASH v1.2.7, QIIME 1.9.1, metaX, Prism GraphPad software v 8.0

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No statistical methods were used to determine sample size. The sample sizes were selected based on previous experience and published literature. For in vitro and in vivo experiments, the minimum of sample size was 3 in all the cases. The sample size is stated in the figure legends.
Data exclusions	No data exclusion was performed.
Replication	All the experiments were replicated. The number of replicates is stated in figure legends.
Randomization	Animals of the indicated ages were randomized in the described groups. For all in vitro experiments, experimental treatment group allocation was random.
Blinding	Histology analyses were performed blinded and the investigators were blinded to group allocation during data collection. For other experiments, the investigators were not blinded to group allocation during the data collection, but were blinded to data analysis.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☐	☒ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	Rabbit monoclonal anti-MyD88 (D80F5), Cell Signaling Technology, Cat# 4283, 1/1000 dilution. Phospho-NF-κB p65 (Ser536) (93H1), Cell Signaling Technology, Cat#3033, 1/1000 dilution. Rabbit polyclonal anti-IL-1β, ABclonal, Cat# A1112, 1/1000 dilution. CYP7A1 Polyclonal Antibody, Thermo, PA5-100892, 1/1000 dilution. CYP8B1 Polyclonal Antibody, Thermo, PA5-37088, 1/1000 dilution. CYP27A1 Monoclonal Antibody(3G3F7), Proteintech, 67045-1-Ig, 1/1000 dilution. CYP7B1 Polyclonal Antibody, Proteintech, 24889-1-AP, 1/1000 dilution. BAAT Polyclonal Antibody, Thermo, PA5-103658, 1/1000 dilution. SLC27A5 Monoclonal (4B11C10), Thermo, MA5-17175, 1/1000 dilution. Rabbit monoclonal (EPR6043) to IKK beta, Abcam, ab124957, 1/1000 dilution. Rabbit monoclonal [EPR21711] to IL-6, Abcam, ab233706,1/1000 dilution. Rabbit monoclonal [E381] to NFκB p105 / p50, Abcam, ab32360, 1/1000 dilution. Rabbit monoclonal [E379] to NF-κB p65, Abcam, ab32536, 1/1000 dilution. Rabbit monoclonal [EP591Y] to TRAF6, Abcam, ab33915, 1/2000 dilution. Mouse monoclonal [DM1A] to alpha Tubulin, Abcam, ab7291, 1/5000 dilution. Rabbit monoclonal [EPR23094-1] to NLRP3, Abcam, ab263899, 1/1000 dilution. Rabbit monoclonal [EPR19672] to Caspase-1, Abcam, ab207802,1/1000 dilution.
Validation	Phospho-NF-κB p65 (Ser536) (93H1), Akira, S. (2003) J Biol Chem 278, 38105-8. Phospho-NF-κB p65 (Ser536) (93H1),Haskill, S. et al. (1991) Cell 65, 1281-9. CYP8B1 Polyclonal Antibody, Nikolaos Nikolaou. et al. (2020) J Endocrinol.245(2),207-218.

CYP27A1 Monoclonal Antibody(3G3F7), Xiaoxi Li. et al. (2022) J Ethnopharmacol.294:115333.
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 Rabbit monoclonal [EPR19672] to Caspase-1, Transl Psychiatry 11:16 (2021).
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 Rabbit monoclonal [EPR6043] to IKK beta,EBioMedicine 65:103283 (2021).
 Rabbit monoclonal [EPR21711] to IL-6, Exp Ther Med 21:568 (2021).
 Rabbit monoclonal [E381] to NFkB p105 / p50, J Mol Neurosci 71:101-111 (2021).
 Rabbit monoclonal [E379] to NF-kB p65, Cancer Gene Ther 28:126-140 (2021).
 Rabbit monoclonal [EP591Y] to TRAF6, Front Immunol 12:637659 (2021).
 Mouse monoclonal [DM1A] to alpha Tubulin, Genes Cells 26:426-446 (2021).

Eukaryotic cell lines

Policy information about [cell lines](#)

Cell line source(s)	THP-1 cells and Vero cells were purchased from ATCC, Huh-7 cells were obtained from Stem Cell Bank, Chinese Academy of Sciences.
Authentication	Huh-7 cells were obtained from Stem Cell Bank, Chinese Academy of Sciences. The serial number is SCSP-526. STR: D5S818: 12; D13S317: 10,11; D7S820: 11; D16S539: 10; vWA: 16,18; TH01: 7; Amelogenin: X; TPOX: 8,11; CSF1PO: 11. THP-1 cells were purchased from ATCC. No. TBI-202. STR: Amelogenin: X,Y; CSF1PO: 11,13; D13S317: 13; D16S539: 11,12; D5S818: 11,12; D7S820: 10;TH01: 8,9,3; TPOX: 8,11; vWA: 16. Vero cells were purchased from ATCC. No. CCL-81.
Mycoplasma contamination	All the Cell lines were tested for mycoplasma contamination by PCR. All the results were negative.
Commonly misidentified lines (See ICLAC register)	None

Animals and other organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research

Laboratory animals	Six-week-old, female C57BL/6 mice purchased from the Model Animal Research Center of Nanjing University (Nanjing, China). Germ-Free mice experiments were carried out at Laboratory Animal Center, Huazhong Agricultural University (Six-week-old, female C57BL/6). Six-week-old, female C57BL/6 mice were treated with an antibiotic cocktail containing ampicillin, neomycin, metronidazole and vancomycin and then referred to as the Abx mice.
Wild animals	No wild animals were used in the study.
Field-collected samples	No field collected samples were used in the study.
Ethics oversight	All animal experiments were strictly carried out in accordance with protocols approved (No.117113) by the Institutional Animal Care and Use Committee of Zhejiang University.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Human research participants

Policy information about [studies involving human research participants](#)

Population characteristics	The median age the 260 patients with SFTS was 62 (IQR 53 to 70) years, and 45% were males. We collected fecal and serum samples at acute and convalescent phases of patients with SFTS.
Recruitment	This is a retrospective clinical study. Adult SFTS patients (aged ≥18 years) who sought treatment at the 990th Hospital in Xinyang city of Henan Province from May 2018 to August 2019 and had fecal samples available, were included in the study. Adult febrile patients without SFTSV infection who sought treatment at the 990th Hospital during the same period and had fecal samples collected were included in the study. Patients with tumours, tuberculosis, diabetes, or other infections (i.e., hepatitis virus, dengue virus, Rickettsia, and Borrelia) were excluded. Healthy individuals determined by routine healthy examination and had fecal samples available were included in the study. The median age, female frequency, and interval time from onset to admission of the SFTS patients included in the study were all comparable with the total population (using data from our previously study: Lancet Infect Dis. 2018; 18:1127-1137).
Ethics oversight	The study was performed with the approval of the Ethical Committee of Beijing Institute of Microbiology and Epidemiology.

Note that full information on the approval of the study protocol must also be provided in the manuscript.